ASYNCHRONOUS STATE ESTIMATION AND LYAPUNOV-CONSTRAINED PREDICTIVE TRACKING FOR A VOLTAGE-LIMITED DIFFERENTIAL-DRIVE ROBOT

QUADE MAIELLO

Abstract. We give a unified mathematical treatment of localization and trajectory tracking for a differential-drive robot whose wheel encoders, rate gyroscopes and time-of-flight range sensors are sampled asynchronously and whose motors are limited in voltage. For estimation, we derive the exact pose update on SE(2), bound the position error caused by clock skew between sensors, and show that the Jacobian of a range measurement against a planar wall is, up to a scalar, the planar wrench of a unit force along the wall normal at the point where the beam lands. It follows that the pose is locally unobservable from range data exactly when those normal lines are concurrent or parallel. For control, we prove that the RAMSETE tracking law has a strict Lyapunov function whose decay rate is uniform in distance traveled, give the exact threshold on its cross-term coefficient, show that the certified rate agrees with the rate of the linearization to three digits, and obtain a computer-assisted bound for the gains used. We then formulate model-predictive tracking with wheel voltages as the decision variables, so that the constraint set is a box independent of the state. We prove that block principal pivoting with a least-index fallback solves the resulting quadratic program exactly in finitely many steps. We show that a time reparameterization of the reference preserves the drawn path under saturation, and that a closed-form projection enforces the Lyapunov decrease on the first planned step, so that the closed loop is locally input-to-state stable with respect to actuator mismatch. Numerical experiments on a physics simulation, scored against the true pose, compare the design against RAMSETE and LTV-LQR tracking across eight robot conditions, and we test how far the comparison depends on the simulator's assumptions. Instruction counts for the target processor, taken under emulation, show that the controller is inexpensive but that the full estimator may exceed its time budget. All results in this paper are simulated; none has yet been checked on hardware.

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1. Introduction

A wheeled robot that must reach a pose to within a centimeter has two problems. It must know where it is, from sensors that disagree in time and are individually biased, and it must get where it is going, with actuators that saturate and a kinematic constraint that forbids sideways motion. The two problems are usually treated apart. In this paper we treat them in one framework for a robot of the class used in the VEX V5 competition: a differential drive of about 7 kg with six brushed motors of limited voltage and current, tracking wheels, up to three rate gyroscopes, and up to six time-of-flight range sensors reading the walls of a field whose interior is about 3.6 m square. Each sensor refreshes on its own clock, between 5 and 33 ms, and the on-board processor is a single Cortex-A9 that must finish the control computation within a 10 ms cycle. That last constraint shapes much of what follows.

1.1. Contributions.

The main results are these.

(1) *The geometry of range sensing* (Section 4.4). Censi [11] showed that the Fisher information of range-finder localization is determined by the expected readings and the orientation of the surfaces at the sensed points. We write one row of the measurement Jacobian as the wrench of a unit force along the wall normal at the hit point, divided by the cosine of the angle of incidence (Proposition 4.4), and read its rank from the line geometry of those normals: an infinitesimal motion is invisible to a set of beams if and only if it is a rotation about a point common to all their normal lines, or a translation orthogonal to all of them (Theorem 4.5). This is the classical construction of Reuleaux for planar immobilization, applied to sensing.

(2) *An explicit strict Lyapunov function for RAMSETE* (Section 6.2). The classical function for this law is not strict: its derivative vanishes on the whole of the cross-track axis, and convergence rests on an invariance argument. Adding a cross term is a standard way to make such a function strict [29], and strict functions for nonholonomic and curve tracking already exist [27, 30]. For this particular law we find that the decay is uniform in distance traveled rather than time, give the exact threshold on the coefficient of the cross term (Theorem 6.4), show that the best certified rate matches that of the linearization to within 0.03%, and give a computer-assisted bound at the implemented gains (Proposition 6.7).

(3) *A predictive tracker with a box constraint set* (Section 7). With voltages as decision variables the input limits form a box, which makes every state feasible and makes an exact active-set solver practical. We give a self-contained proof of finite termination for block principal pivoting with a least-index fallback on strictly convex box quadratic programs (Theorem 7.4), adapting the induction of Murty [34] to two-sided bounds. We prove a mirror symmetry of the terminal Riccati cost (Proposition 7.2) and give a closed form for its divergence as speed tends to zero (Proposition 6.8).

(4) *A reference clock* (Section 7.5). In the spirit of on-line time scaling [13], a scalar clock rate, set by the voltage the next half second will need and by how far behind the robot has fallen, keeps the reference reachable. The drawn path is preserved exactly, and only its timing changes. We give the steady lag in closed form (Proposition 7.8).

(5) *A Lyapunov guard* (Section 7.6). A variant of Lyapunov-based predictive control [33]: a single closed-form projection forces the plan's first step to shrink the strict function at a fixed fraction of the rate RAMSETE would achieve. This yields a local input-to-state stability estimate with respect to the mismatch between planned and achieved wheel speeds, with an explicit ultimate bound (Theorem 7.10). Remark 7.11 states what this does not cover.

(6) *Cost on the target processor* (Section 9.7). Instruction counts for the compiled code on an emulated Cortex-A9 show that the estimator, and not the controller, is the main threat to the real-time budget.

We also collect in one place, with derivations, several results that are known or elementary but are not usually stated together: a clock-skew bound (Proposition 4.1), the selection bias that arises from stale readings (Proposition 4.3), the effect of unmodeled biases on filter consistency (Proposition 4.8), and the lattice structure of feasible speed profiles (Theorem 5.1). We make no claim of priority for any individual result. The novelty, if any, lies in the combination and in the explicit constants, and the survey below is not exhaustive.

1.2. **Related work.** Tracking laws for the unicycle go back to Kanayama et al. [23] and Samson [37]. The form used here is the one in De Luca, Oriolo and Vendittelli [14], and it takes the name RAMSETE from the volume in which that survey appears. Jiang and Nijmeijer [20] use backstepping and obtain global results. Brockett's theorem [9] excludes smooth static stabilization of a pose, and Aicardi et al. [1] avoid it with polar coordinates. On strict Lyapunov functions in general we refer to Malisoff and Mazenc [29]. Strict functions for nonholonomic tracking under persistency of excitation are constructed by Maghenem, Loría and Panteley [27], with a general tool in Maghenem and Loría [28]. Malisoff, Mazenc and Zhang [30] build one for curve tracking, and Loría et al. [26] give the theory of uniform convergence behind such arguments.

Model-predictive control is surveyed by Mayne et al. [32] and Rawlings, Mayne and Diehl [35]. Lyapunov-based predictive control, in which a known stabilizing law supplies a constraint on the plan, is due to Mhaskar, El-Farra and Christofides [33]. Their constraint is imposed inside the optimization; ours is a projection applied afterward, which leaves the quadratic program a pure box problem. Active-set solvers for embedded use are represented by qpOASES [16]. For box constraints, the primal-dual active set method of Kunisch and Rendl [24] and Hintermüller, Ito and Kunisch [18] and the block principal pivoting of Júdice and Pires [21] are close relatives of the solver in Section 7.4, and the least-index safeguard is due to Murty [34]. Cimini and Bemporad [12] certify the worst-case iteration count of the Kunisch–Rendl method for the parametric box problems of linear model-predictive control, and they note that its convergence is guaranteed only under restrictive conditions. That method can cycle, which is why Algorithm 1 carries a fallback.

Time-optimal parameterization along a path originates with Bobrow, Dubowsky and Gibson [8] and Shin and McKay [38], with convex formulations by Verscheure et al. [42] and Lipp and Boyd [25]. On-line time scaling under torque limits is due to Hollerbach [19] and Dahl and Nielsen [13], and our clock belongs to that line of work.

On the estimation side we rely on Kalman [22], Anderson and Moore [2], Bar-Shalom, Li and Kirubarajan [6], and Barfoot [7]. Late measurements are treated by Bar-Shalom [5]. Monte Carlo localization follows Thrun et al. [41, 40] and Arulampalam et al. [4]. The dependence of range-finder Fisher information on surface orientation is due to Censi [11], and degeneracy in scan matching is commonly detected from the spectrum of the information matrix [45]. The observability rank condition is that of Hermann and Krener [17], and form closure is treated by Mason [31].

1.3. **Organization and notation.** Section 2 fixes the geometry and Section 3 derives the actuator model. Section 4 treats localization and Section 5 builds the reference. Section 6 studies tracking laws in error coordinates, and Section 7 builds the predictive tracker on them. Section 8 joins the two halves, and Section 9 reports the experiments.

Throughout, $\mathrm{R}(a)$ is rotation by a and $\mathrm{J} = \mathrm{R}(\pi/2)$. For $a, b \in \mathbb{R}^2$ we write $a \times b = a_1 b_2 - a_2 b_1 = a^{\mathsf{T}} \mathrm{J}^{\mathsf{T}} b$. We put $\mathbf{e}(a) = (\cos a, \sin a)^{\mathsf{T}}$, $\operatorname{sinc} x = \sin x / x$ with $\operatorname{sinc} 0 = 1$, and $[x]_+ = \max(x, 0)$. We write $\operatorname{clip}(x, a, b) = \min(\max(x, a), b)$. Headings are real numbers, not angles modulo 2π, unless they are explicitly wrapped. All quantities are in SI units. The symbol ℓ always denotes the track width.

2. Geometry and kinematics

A pose is $g = (p, \theta) \in \mathrm{SE}(2)$ with $p \in \mathbb{R}^2$. Composition is $(p_1, \theta_1)(p_2, \theta_2) = (p_1 + \mathrm{R}(\theta_1) p_2,\ \theta_1 + \theta_2)$. A body twist is a triple $\xi = (u, \omega) \in \mathbb{R}^2 \times \mathbb{R}$, and a trajectory of the rigid body satisfies

$$\dot{p} = \mathrm{R}(\theta)\, u, \qquad \dot{\theta} = \omega. \tag{2.1}$$

Lemma 2.1 (Exponential map)**.** *If ξ is constant on $[0, 1]$ then $g(1) = g(0) \exp \xi$, where*

$$\exp(u, \omega) = \big(\mathrm{V}(\omega)\, u,\ \omega\big), \qquad \mathrm{V}(\omega) = \int_0^1 \mathrm{R}(\omega t)\, \mathrm{d}t = \operatorname{sinc}(\omega/2)\, \mathrm{R}(\omega/2). \tag{2.2}$$

Proof. Integrating (2.1) with $\theta(t) = \theta(0) + \omega t$ gives $p(1) - p(0) = \mathrm{R}(\theta(0)) \int_0^1 \mathrm{R}(\omega t)\, \mathrm{d}t\ u$. The entries of the integral are $\sin\omega/\omega$ and $(1 - \cos\omega)/\omega$, and the half-angle identities $\sin\omega = 2 \sin\frac{\omega}{2} \cos\frac{\omega}{2}$ and $1 - \cos\omega = 2\sin^2\frac{\omega}{2}$ give the product form. □

Lemma 2.1 shows at once how two common approximations err.

Proposition 2.2. *Let an increment (Δ, ϕ) be measured over one step, with $\Delta \in \mathbb{R}^2$ in the body frame at the start of the step. Relative to the exact update $\mathrm{V}(\phi)\Delta$, the forward Euler update Δ has error norm $\|\Delta\|\,(|\phi|/2 + O(\phi^3))$, and the midpoint update $\mathrm{R}(\phi/2)\Delta$ has error norm $\|\Delta\|\,(1 - \operatorname{sinc}(\phi/2)) = \|\Delta\|\,(\phi^2/24 + O(\phi^4))$.*

Proof. By (2.2) we have $\mathrm{V}(\phi) - \mathrm{R}(\phi/2) = (\operatorname{sinc}(\phi/2) - 1)\mathrm{R}(\phi/2)$. For Euler, write $\mathrm{V}(\phi) = aI + c\mathrm{J}$ with $a = \operatorname{sinc}\phi$ and $c = (1 - \cos\phi)/\phi$. Then $I - \mathrm{V}$ is a scalar multiple of a rotation, of norm $((1 - a)^2 + c^2)^{1/2} = |\phi|/2 + O(\phi^3)$. □

So the midpoint rule, which is what "rotate by the average heading" means, already has the right direction, and it errs only in the length of the chord against the arc.

2.1. The differential drive.

A differential drive has $u = (v, 0)^{\mathsf{T}}$, so that (2.1) becomes the unicycle

$$\dot{x} = v \cos\theta, \qquad \dot{y} = v \sin\theta, \qquad \dot{\theta} = \omega. \tag{2.3}$$

With left and right wheel surface speeds v_L, v_R and effective track width ℓ,

$$v = \tfrac{1}{2}(v_L + v_R), \qquad \omega = (v_R - v_L)/\ell. \tag{2.4}$$

The width ℓ is a fitted constant and exceeds the geometric width, because the tires scrub in a turn.

The vector fields of (2.3) and their Lie bracket span the tangent space, so every pose is reachable, but sideways motion is available only through the bracket. The next bound makes this quantitative and will explain the behavior at the ends of paths in Section 8.

Proposition 2.3 (Lateral reach)**.** *Let n be the unit normal to the direction $\mathbf{e}(\theta_0)$, and suppose that $|\theta(t) - \theta_0| \le \psi < \pi/2$ on $[0, T]$. Then*

$$\big|n^{\mathsf{T}}(p(T) - p(0))\big| \le \sin\psi \int_0^T |v(t)|\, \mathrm{d}t.$$

In particular, moving a robot sideways by d while keeping its heading within ψ of θ_0 requires a path of length at least $d/\sin\psi$.

Proof. By (2.3), $n^{\mathsf{T}}\dot{p} = v\sin(\theta - \theta_0)$. Integrate. □

3. Actuator model

Let each side have N motors with resistance R, back-EMF constant K_e, torque constant K_t, viscous coefficient β and Coulomb friction τ_c, geared by G motor turns per wheel turn to wheels of radius r_w with reflected inertia J_s. Let m be the robot's mass and μ_r the rolling-resistance coefficient. For a voltage u the current in one motor is $i = (u - K_e G v/r_w)/R$, subject to a limit $|i| \le i_{\max}$, and the motion of one side, carrying half the mass, obeys

$$\Big(\frac{m}{2} + \frac{J_s}{r_w^2}\Big)\dot{v} = \frac{NG}{r_w}\Big(K_t i - \tau_c \operatorname{sgn} v - \beta\frac{Gv}{r_w}\Big) - \mu_r \frac{mg}{2}\operatorname{sgn} v. \tag{3.1}$$

Proposition 3.1. *While the current limit is inactive,* (3.1) *is equivalent to*

$$u = k_S \operatorname{sgn} v + k_V v + k_A \dot{v}, \tag{3.2}$$

with

$$k_V = \frac{G}{r_w}\Big(K_e + \frac{R\beta}{K_t}\Big), \quad k_A = \frac{R}{K_t}\,\frac{r_w m/2 + J_s/r_w}{NG}, \quad k_S = \frac{R}{K_t}\Big(\tau_c + \frac{r_w \mu_r m g}{2NG}\Big).$$

Proof. Multiply (3.1) by $r_w R/(NGK_t)$, substitute $Ri = u - K_e G v/r_w$, and collect terms. □

Treating the two sides independently is a modeling choice, and the next result says when it is exact.

Proposition 3.2 (Decoupling)**.** *Let F_L, F_R be the ground forces on the two sides and I the yaw inertia, so that $m\dot{v} = F_L + F_R$ and $I\dot{\omega} = (F_R - F_L)\ell/2$. Then*

$$\dot{v}_R = \Big(\frac{1}{m} + \frac{\ell^2}{4I}\Big)F_R + \Big(\frac{1}{m} - \frac{\ell^2}{4I}\Big)F_L,$$

and symmetrically for $\dot{v}_L$. The sides are dynamically independent if and only if $4I = m\ell^2$.

Proof. Use $v_R = v + \omega\ell/2$ and substitute. □

For the simulated robot of Section 9, $m = 6.8$ kg, $I = 0.20\ \mathrm{kg\,m^2}$ and $\ell = 0.33$ m, so the cross coefficient is less than 4% of the direct one. We take the sides as independent.

Let a wheel follow a reference speed v_r and write $\xi = v - v_r$. When $\operatorname{sgn} v = \operatorname{sgn} v_r$, subtracting (3.2) at v_r from (3.2) at v gives

$$\dot{\xi} = \frac{\delta - k_V \xi}{k_A}, \qquad \delta = u - (k_S \operatorname{sgn} v_r + k_V v_r + k_A \dot{v}_r). \tag{3.3}$$

Let $T_w = k_A/k_V$.

Lemma 3.3 (Exact sampling)**.** *If δ is constant on $[0, h]$ then*

$$\xi(h) = \alpha\,\xi(0) + \gamma\,\delta, \qquad \frac{1}{h}\int_0^h \xi\,\mathrm{d}t = a_1\xi(0) + a_2\delta, \tag{3.4}$$

where $\alpha = e^{-h/T_w}$, $\gamma = (1-\alpha)/k_V$, $a_1 = (1-\alpha)T_w/h$, and $a_2 = (h - (1-\alpha)T_w)/(k_V h)$.

Proof. The solution of (3.3) is $\xi(t) = \xi(0)e^{-t/T_w} + (\delta/k_V)(1 - e^{-t/T_w})$. Evaluate at h, and integrate over $[0, h]$. □

4. Localization

4.1. **State and motion model.** The estimator keeps the pose together with the calibration quantities that would otherwise appear as unmodeled bias:

$$x = (p, \theta, b_g, s_g, \mu_\parallel, \mu_\perp, o_\parallel, o_\perp, d_1, \dots, d_m) \in \mathbb{R}^{9+m}.$$

Here b_g and s_g are the gyro bias and scale error, $\mu_\parallel, \mu_\perp$ are relative radius errors of the tracking wheels, $o_\parallel, o_\perp$ are their mounting offsets, and d_i is the bias of range sensor i. A parallel wheel, mounted at lateral offset $o_\parallel$, rolls along the body x-axis, and a perpendicular wheel, mounted at forward offset $o_\perp$, rolls along the body y-axis. Over a step of length h let the wheels turn through $\Delta\phi_\parallel$ and $\Delta\phi_\perp$ and the gyros report $\Delta\Gamma$, the mean over n units. The velocity of a wheel's contact point in the body frame gives

$$\begin{gathered} \Delta\theta = \frac{\Delta\Gamma - b_g h}{1 + s_g}, \\ \Delta x = (1+\mu_\parallel) r_\parallel \Delta\phi_\parallel + o_\parallel \Delta\theta, \qquad \Delta y = (1+\mu_\perp) r_\perp \Delta\phi_\perp - o_\perp \Delta\theta. \end{gathered} \tag{4.1}$$

By Lemma 2.1, if the body twist is constant over the step then the pose update is exactly

$$(p, \theta) \mapsto (p, \theta) \exp\big((\Delta x, \Delta y), \Delta\theta\big), \tag{4.2}$$

and the other states are unchanged. The Jacobians of this map are in Appendix A.

4.2. **Time alignment.** The channels of (4.1) refresh at different rates and phases. Combining the newest value of each would mix samples up to 10 ms apart. The estimator instead stamps every sample and, at each tick, computes the *horizon* $t^\star = \min_c t_c$, the earliest of the latest sample times over all channels c. It then evaluates every cumulative channel at $t^\star$ by linear interpolation and forms the increments of (4.1) between consecutive horizons. Two results explain why this matters.

Proposition 4.1 (Clock skew)**.** *Let a robot move with v, ω continuous and $\theta \in C^2$, with θ extended to $[-\delta, 0]$, and let its path be integrated with a heading delayed by δ, so that $\hat{p}(0) = p(0)$ and $\dot{\hat{p}} = v\,\mathbf{e}(\theta(t-\delta))$. Then*

$$\hat{p}(T) - p(T) = -\delta \int_0^T v\,\omega\,\mathbf{J}\mathbf{e}(\theta)\,\mathrm{d}t + O(\delta^2), \tag{4.3}$$

so that $\|\hat{p}(T) - p(T)\| \le \delta \int_0^T |v\omega|\,\mathrm{d}t + O(\delta^2)$. If ω is constant, then $\hat{p}(T) - p(0) = \mathrm{R}(-\omega\delta)(p(T) - p(0))$ exactly, and the error is $2|\sin(\omega\delta/2)|\,\|p(T) - p(0)\|$.

Proof. A Taylor expansion gives $\mathbf{e}(\theta(t-\delta)) = \mathbf{e}(\theta) - \delta\,\omega\,\mathbf{J}\mathbf{e}(\theta) + O(\delta^2)$; integrate against v. For constant ω, $\theta(t-\delta) = \theta(t) - \omega\delta$, so $\mathbf{e}(\theta(t-\delta)) = \mathrm{R}(-\omega\delta)\mathbf{e}(\theta(t))$. Pulling the fixed rotation outside the integral gives the second claim. □

A skew of 10 ms during a turn at 3 rad/s rotates the whole displacement by 1.7°, which is 3 cm over a meter. The error depends on δ to first order, while interpolation removes it to second order:

Lemma 4.2. *If $q \in C^2$ and Iq is the linear interpolant of q between samples h apart, then $|q - Iq| \le \frac{h^2}{8} \sup |\ddot{q}|$.*

This is the standard interpolation bound.

A subtler failure comes from reading a device faster than it refreshes. Optical encoders and gyros refresh every 10 ms and are read every 5 ms, and the time of a reading is known only as the time it was read. A repeated value then looks like a stopped channel, and the next new value looks like a jump.

Proposition 4.3 (Selection from stale data)**.** *Let two redundant wheels measure the same constant-speed motion of Δ per step. Wheel A updates every step. Wheel B updates every p steps, so that it reports 0 for $p-1$ steps and then $p\Delta$. Suppose a fusion rule treats a disagreement as a wheel lifting off the ground and keeps the larger reading. Then the fused distance over a period is $(2p-1)\Delta$ instead of $p\Delta$, which is a ratio of $2-1/p$. If instead the two wheels are compared over a window of K steps, the discrepancy is at most $(p-1)\Delta$, and is zero when p divides K.*

Proof. The maximum of the two per-step readings is Δ on $p-1$ steps and $p\Delta$ on one, which proves the first claim. In a window of K steps, wheel A reports $K\Delta$ and wheel B reports $c\,p\Delta$ with $c \in \{\lfloor K/p \rfloor, \lceil K/p \rceil\}$, and $|K - cp| \le p-1$. □

For $p = 2$ the estimate runs 1.5 times faster than the robot, and this factor was observed in simulation before the following rules were adopted. The implementation does two things. A repeated reading is dropped until it has held for 12 ms, longer than any device takes to refresh, after which the channel has really stopped. And the redundant wheels are compared over a window of $K = 10$ steps, with a threshold on $|\Sigma_A - \Sigma_B|$ of $0.002 + 0.1 \max(|\Sigma_A|, |\Sigma_B|) + 0.02\,\Sigma|\Delta\theta|$ meters.

When the pair of tracking wheels both leave the ground, comparing them with each other cannot detect it. The mean drive-encoder travel then gives a third witness. If it exceeds 2 mm per step while the tracking wheel reports less than 0.3 of it, then $\Delta\phi_\parallel$ is replaced by the drive travel with an inflated variance.

4.3. **Prediction.** The filter propagates $\hat{x}^+ = f(\hat{x}, w)$, $P^+ = FPF^\mathsf{T} + GQ_wG^\mathsf{T} + Q_x$, where F, G are the Jacobians of (4.1)–(4.2) with respect to the state and to the increments $w = (\Delta\phi_\parallel, \Delta\phi_\perp, \Delta\Gamma)$. The increment noise is

$$Q_w = \operatorname{diag}\Big(\frac{q_{\text{slip}}|r_\parallel \Delta\phi_\parallel|}{r_\parallel^2},\ \frac{q_{\text{slip}}|r_\perp \Delta\phi_\perp|}{r_\perp^2},\ \frac{N_g^2 h + 2q_g^2}{n}\Big).$$

The first two entries model slip whose variance grows with distance, at rate q_{slip}. In the third, N_g is the angle random walk of one gyro and q_g the error of one cumulative reading, of which an increment contains two. The calibration states perform slow random walks, collected in Q_x.

When no channel has changed for 0.12 s, the robot is declared stationary. Then $\Delta\theta$ is set to zero, and $\Delta\Gamma/h$ is used as a direct measurement of b_g with variance $N_g^2/(nh) + 4q_g^2/(nh^2)$.

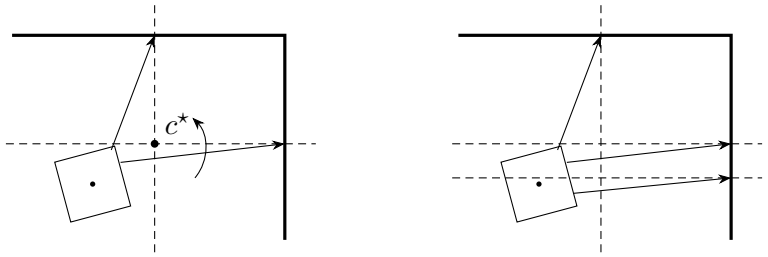
4.4. **Range sensing against planar walls.** A range sensor sits at c in the body frame and points along body angle φ, so its beam starts at $o = p + \mathrm{R}(\theta)c$ with direction $d = \mathbf{e}(\theta + \varphi)$. Let a wall be the line $\{q : n^\mathsf{T} q = c_0\}$ with unit normal n pointing out of the field, so that $n^\mathsf{T} d > 0$ for a beam that reaches it. The predicted range is

$$\rho(p, \theta) = \frac{c_0 - n^\mathsf{T} o}{n^\mathsf{T} d}. \tag{4.4}$$

Proposition 4.4 (The wrench form)**.** *Let $q = o + \rho d$ be the hit point and $a = q - p$ the lever arm from the robot's reference point. Then*

$$\frac{\partial \rho}{\partial (p, \theta)} = -\frac{1}{n^\mathsf{T} d}\begin{pmatrix} n^\mathsf{T} & a \times n \end{pmatrix}. \tag{4.5}$$

Proof. Consider an infinitesimal twist $(\delta p, \delta\theta)$ of the robot, with the ray rigidly attached. The material point of the ray at parameter ρ moves by $\delta q = \delta p + \delta\theta\, \mathrm{J}a$, and the direction of the ray changes by a first-order amount. The new hit point is $q + \delta q + \delta\rho\, d$ plus terms of second order, and it must lie on the wall, so that $n^\mathsf{T}\delta q + \delta\rho\, n^\mathsf{T} d = 0$. Since $n^\mathsf{T}\mathrm{J}a = a \times n$, this gives (4.5). □



(a) rank 2 (b) rank 3

FIGURE 1. Beams (solid arrows) and their wall normals (dashed). In (a) the two normals meet at $c^\star$, and a small rotation of the robot about $c^\star$ changes neither range to first order. A second beam on the same wall in (b) removes the common point.

The row vector $(n^\mathsf{T},\ a \times n)$ in (4.5) is the planar wrench of a unit force along n applied at q, and $n^\mathsf{T} d$ is the cosine of the incidence angle. A sensor thus measures the component of the robot's motion along the wall normal at the hit point, and (4.5) is the single-beam form of the dependence on surface orientation found by Censi [11]. Note that the direction of the beam enters only through the cosine: two beams that land at the same point of the same wall give proportional rows.

Let $L_i = \{q_i + t n_i : t \in \mathbb{R}\}$ be the normal line at the ith hit point, and let $H \in \mathbb{R}^{k\times 3}$ collect the rows (4.5).

Theorem 4.5 (Invisible motions)**.** *A nonzero twist $(\delta p, \delta\theta)$ lies in* $\ker H$ *if and only if one of the following holds:*

(1) *$\delta\theta \neq 0$, and the center of rotation $c^\star = p + \mathrm{J}\delta p/\delta\theta$ lies on every L_i;*
(2) *$\delta\theta = 0$, and δp is orthogonal to every n_i.*

Consequently $\operatorname{rank} H = 3$ *if and only if the lines L_i are neither all concurrent nor all parallel.*

Proof. The ith row annihilates the twist if and only if $n_i^\mathsf{T}(\delta p + \delta\theta\, \mathrm{J} a_i) = 0$. That is, the material point q_i has no velocity along n_i. If $\delta\theta \neq 0$, the velocity field of the twist is $x \mapsto \delta\theta\, \mathrm{J}(x - c^\star)$, because $\delta p + \delta\theta \mathrm{J}(c^\star - p) = \delta p + \mathrm{J}^2 \delta p = 0$. So the condition reads $n_i^\mathsf{T} \mathrm{J}(q_i - c^\star) = 0$, which says that $q_i - c^\star$ is parallel to n_i, or $c^\star \in L_i$. If $\delta\theta = 0$, the condition is $n_i^\mathsf{T}\delta p = 0$. A nonzero δp orthogonal to all n_i exists if and only if the n_i are all parallel. For the rank claim, $\ker H$ is nonzero if and only if (1) or (2) holds for some twist. Condition (1) is solvable exactly when the lines have a common point, and condition (2) exactly when they are all parallel. □

This is the construction of Reuleaux for planar form closure [31]: a planar body is immobilized to first order by frictionless contacts when their normals have no common point. Here each beam acts as a contact. Figure 1 shows the two cases. In scan matching, degeneracy is usually detected numerically from the smallest eigenvalue of the information matrix [45]; for planar walls the theorem names the invisible motion directly.

Corollary 4.6 (Rectangular field)**.** *In a rectangular field all normal lines are horizontal or vertical. Then* $\operatorname{rank} H = 3$ *if and only if there are at least two distinct lines in one family and at least one line in the other. If the heading is known from the gyros, position is determined by any two beams on perpendicular walls.*

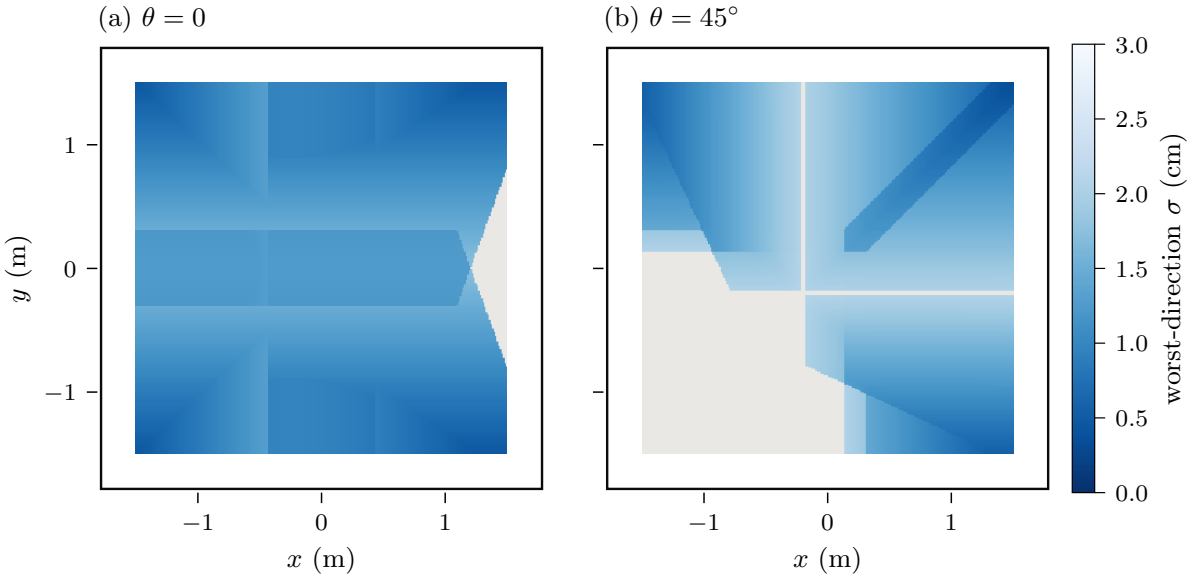


FIGURE 2. Cramér–Rao bound for position from the range sensors alone, by robot position, with a heading prior of 0.2°. Gray regions have fewer than two usable beams on perpendicular walls, so one direction of position is unobservable from range data (Corollary 4.6). The usable fraction of the field is 97% at $\theta = 0$ and 75% at $\theta = 45°$, with median bounds of 1.14 cm and 1.52 cm.

Proof. A set of horizontal and vertical lines is concurrent or parallel exactly when it is all horizontal, all vertical, or consists of one line of each family. The second claim is Theorem 4.5(2) with $\delta\theta = 0$ imposed: two perpendicular normals leave no orthogonal δp. □

With independent noise of standard deviation σ_i the Fisher information of the pose is $\mathcal{I} = \sum_i H_i^{\mathsf{T}} H_i / \sigma_i^2$, and $\mathcal{I}^{-1}$ bounds the covariance of any unbiased estimator from below. Figure 2 maps the worst-direction position uncertainty for the five-sensor layout of Section 9, with a heading prior of 0.2°, namely the reciprocal square root of the smallest eigenvalue of the Schur complement of $\mathcal{I}$ onto position. The discontinuities are the lines where a beam changes walls or passes the 50° incidence or 2 m range limit.

The implementation refines (4.4) in three ways. A real time-of-flight sensor reports the nearest return in its field of view, so the predicted range is the smallest of three rays, at the center and the edges of a cone of half-angle 0.06 rad. Near a wall this function has kinks, and a reading is rejected if the two-sided difference used for the Jacobian jumps by more than 2 cm, or if the beam lands within 3 cm of a segment end, or beyond 50° incidence or 2 m range. The measurement variance is

$$\sigma^2 = \Big(\frac{0.004 + 0.008z}{\max(\cos\iota,\, 0.5)}\Big)^2 + (0.003\,\|\dot{p}\|)^2 + (0.002z)^2 + 10^{-6}, \tag{4.6}$$

where z is the calibrated range and ι the incidence angle. This is further inflated when the sensor reports low confidence. The innovation includes the bias state, $\nu = z - \rho - d_i$.

4.5. **Measurement updates.** All updates are scalar, in Joseph form:

$$K = \frac{Ph}{s}, \quad s = h^{\mathsf{T}} P h + \sigma^2, \quad \hat{x}^+ = \hat{x} + K\nu, \quad P^+ = (I - Kh^{\mathsf{T}})P(I - Kh^{\mathsf{T}})^{\mathsf{T}} + \sigma^2 KK^{\mathsf{T}}. \tag{4.7}$$

The Joseph form holds for any gain and keeps P^+ symmetric and positive semidefinite in floating point [10]. Processing readings one at a time loses nothing:

Proposition 4.7. *For a linear measurement with independent noise, the sequence of scalar updates* (4.7) *gives the same mean and covariance as the batch update, in any order.*

Proof. In information form, a scalar update adds $h_i h_i^{\mathsf{T}}/\sigma_i^2$ to P^{-1} and $h_i z_i/\sigma_i^2$ to $P^{-1}\hat{x}$. Both operations commute, and their sum over i is the batch update. □

A reading is accepted only if $\nu^2/s \le 9$. Under the filter's own model, ν^2/s is χ^2_1, so this rejects 0.27% of good readings. A beam blocked by an object that is not in the map reads far shorter than predicted and usually fails the test.

4.6. **Why biases must be states.** Filter consistency is measured by the normalized estimation error squared, NEES $= e^{\mathsf{T}} P^{-1} e$, whose mean is the dimension of e when the filter is consistent.

Proposition 4.8 (Unmodeled bias)**.** *Let scalar readings $z_k = \xi + \beta + \eta_k$ have constant bias β and independent noise of variance R. A filter that omits β, with a flat prior, has after N readings*

$$\mathbb{E}[\text{NEES}] = 1 + \frac{N\beta^2}{R},$$

which grows without bound. A filter that carries a bias state with prior variance σ_β^2 has

$$\mathbb{E}[\text{NEES}] = \frac{\beta^2 + R/N}{\sigma_\beta^2 + R/N},$$

which is bounded by $\max(1, \beta^2/\sigma_\beta^2)$.

Proof. In the first case $\hat{\xi} - \xi = \beta + \bar{\eta}$ with reported variance R/N. In the second, ξ and the bias are not separately observable from the readings. The posterior variance of ξ is then $\sigma_\beta^2 + R/N$, and the error $\beta + \bar{\eta}$ is unchanged. □

Calibration residuals are biases and not noise: a 3 mm offset read 50 times at $\sigma = 2$ mm gives a NEES of about 114 in the first case. Removing the calibration states from the full filter shows the same effect (Table 3).

4.7. **Late measurements.** A range reading is available about 25 ms after its capture. At 2 m/s, fusing it with the current pose would misplace it by 5 cm. The estimator stores $(\hat{x}, P)$ at each of the last 0.3 s of steps. When a reading arrives, it attaches the reading to the step containing its capture time, rewinds to that step, and replays all later steps and their measurements.

Proposition 4.9. *Replay gives exactly the estimate of a filter that received all measurements in capture order, provided that each is no older than the stored history. The cost per late reading is at most $\lceil T_h/h \rceil$ prediction steps, or 60 here.*

Proof. The filter is a deterministic function of the ordered sequence of steps and measurements, and replay recomputes that function from a stored state. □

Exact methods that avoid replay exist [5], but at the sizes here replay costs less to implement correctly, and it also re-linearizes.

Between steps, the published pose is extrapolated from the filter time to the present by (4.2) with the latest velocity, so a controller reads an estimate of the current pose.

4.8. **The particle supervisor.** A Gaussian filter that has lost the robot rejects every good reading, and the gate of the previous subsection then keeps it lost. This happens when the robot is shoved with its tracking wheels off the ground. A particle filter with 300 particles runs alongside. It uses the likelihood

$$p(z \mid \rho) = 0.75\,\mathcal{N}(z; \rho, \sigma_\rho^2) + 0.15 \cdot 1.5 e^{-1.5z} \mathbf{1}_{z<\rho} + 0.10/\rho_{\max},$$

with $\sigma_\rho = 0.012 + 0.012\rho$, where the second term accounts for beams blocked short of the wall by objects not in the map. Latency is handled by moving each particle back along the estimated motion since capture. Weights are resampled by the low-variance scheme when the effective sample size $(\sum_i w_i^2)^{-1}$ falls below half the particle count, and recovery particles are injected at a rate set by fast and slow averages of the likelihood, in the manner of augmented Monte Carlo localization [40].

The particle mean replaces the filter's position only under strict conditions: a Mahalanobis distance above 25 between the two, a particle spread under 5 cm, a separation over 10 cm, and, over at least eight consecutive updates, a mean inlier fraction among the fresh readings that exceeds the filter's by 0.25. The last test matters, because a reading blocked by clutter fails for every candidate pose and so cannot favor either side. This logic is a heuristic, and we make no claim about it beyond the experiments in Section 9.

4.9. **Reported uncertainty.**

Proposition 4.10. *Let $e \sim \mathcal{N}(0, P)$ in $\mathbb{R}^2$ and let $\lambda_{\max}$ be the largest eigenvalue of P. Then for $c > 0$, $\Pr(\|e\| \le \sqrt{c\lambda_{\max}}) \ge 1 - e^{-c/2}$.*

Proof. We have $\|e\|^2 \le \lambda_{\max} e^{\mathsf{T}} P^{-1} e$, and $e^{\mathsf{T}} P^{-1} e$ is χ^2_2, whose tail is $e^{-c/2}$. □

The implementation reports the radius with $c = 11.83$, for which the probability is 99.73%, the two-dimensional analogue of three standard deviations.

5. The reference

5.1. **Paths.** A path is a chain of cubic Bézier segments [15]. With control points $P_0, \dots, P_3$,

$$C(u) = (1-u)^3 P_0 + 3(1-u)^2 u P_1 + 3(1-u) u^2 P_2 + u^3 P_3, \qquad u \in [0,1],$$

and the curvature is $\kappa = (C' \times C'')/\|C'\|^3$. At the start of a segment,

$$\kappa(0) = \frac{2}{3} \frac{(P_1 - P_0) \times (P_2 - P_1)}{\|P_1 - P_0\|^3}. \tag{5.1}$$

The first handle is placed along the robot's heading, so the path leaves tangent to it. At interior joints the two handles are collinear, which makes the chain G^1. It is not G^2 in general: (5.1) shows that curvature on the two sides of a joint depends on the handle lengths and the next control point, and it may jump. A jump in curvature is a jump in the turn rate demanded at constant speed, and the tracker of Section 7 anticipates it through its preview.

5.2. **Speed profiles.** Parameterize the path by arc length $s \in [0, S]$ and put $w(s) = v(s)^2$, so that the acceleration is $a = \frac{1}{2} w'$. The constraints take the form $0 \le w \le \bar{w}(s)$ and

$$-2D(s, w) \le w' \le 2A(s, w). \tag{5.2}$$

Here $\bar{w}$ combines a wheel-speed limit $v(1 + |\kappa|\ell/2) \le v_{\max}$, a lateral acceleration limit $v^2|\kappa| \le a_c$, and the steady voltage limit $k_S + k_V v(1 + |\kappa|\ell/2) \le U$. The bounds A, D combine the acceleration limit with the voltage limit on each wheel, where $v_w = v(1 \pm \kappa\ell/2)$ and $a_w = a(1 \pm \kappa\ell/2)$.

Theorem 5.1 (Largest profile)**.** *Let $\bar{w} \geq 0$ be bounded, and let $A, D > 0$ be bounded, measurable in s and continuous in w. (Curvature may jump at the joints of the path, so continuity in s cannot be assumed.) Let $\mathcal{F}$ be the set of Lipschitz functions w on $[0, S]$ with $w(0) \leq w_\mathrm{a}$, $w(S) \leq w_\mathrm{b}$, $0 \leq w \leq \bar{w}$, and (5.2) almost everywhere. If $w_1, w_2 \in \mathcal{F}$ then $\max(w_1, w_2) \in \mathcal{F}$. Moreover $\mathcal{F}$ has a greatest element $w^\star$, and $w^\star$ minimizes the traversal time $\int_0^S w^{-1/2}\,\mathrm{d}s$ over $\mathcal{F}$.*

Proof. Let $m = \max(w_1, w_2)$, which is Lipschitz. On the open set $\{w_1 > w_2\}$ we have $m = w_1$, so (5.2) holds there, and likewise on $\{w_2 > w_1\}$. On $\{w_1 = w_2\}$, the Lipschitz function $w_1 - w_2$ has zero derivative almost everywhere, and so $m' = w_1' = w_2'$ almost everywhere there. So (5.2) holds for m almost everywhere. The other constraints are immediate.

The set $\mathcal{F}$ contains 0. All its members share one Lipschitz constant, by (5.2) and the boundedness of A and D. Let $\{s_k\}$ be dense in $[0, S]$ and choose, for each k, members of $\mathcal{F}$ approaching $\sup_{\mathcal{F}} w(s_k)$. The pointwise maxima of the first n of all these form an increasing sequence $w_n \in \mathcal{F}$, which converges uniformly to $w^\star = \sup \mathcal{F}$. Passing to the limit in $w_n(t) - w_n(r) \leq \int_r^t 2A(\sigma, w_n(\sigma))\,\mathrm{d}\sigma$ by dominated convergence, and likewise for the lower bound, shows that $w^\star \in \mathcal{F}$. Finally, $w \mapsto \int w^{-1/2}$ is decreasing in w pointwise. □

On a grid $s_0 < \cdots < s_n$ with spacing Δ_i and bounds $a_i, d_i > 0$ on each segment, the constraints become difference constraints, and the greatest element takes two sweeps. Define

$$\begin{aligned} w_0^f &= \min(\bar{w}_0, w_\mathrm{a}), \quad w_i^f = \min(\bar{w}_i,\ w_{i-1}^f + 2a_{i-1}\Delta_{i-1}), \\ w_n^b &= \min(w_n^f, w_\mathrm{b}), \quad w_i^b = \min(w_i^f,\ w_{i+1}^b + 2d_i\Delta_i). \end{aligned} \tag{5.3}$$

Proposition 5.2. *The sequence w^b is the greatest with $w_0 \leq w_\mathrm{a}$, $w_n \leq w_\mathrm{b}$, $w_i \leq \bar{w}_i$, $w_{i+1} - w_i \leq 2a_i\Delta_i$ and $w_i - w_{i+1} \leq 2d_i\Delta_i$. With constant acceleration on a segment, the time to traverse it is exactly $2\Delta_i/(v_i + v_{i+1})$.*

Proof. Feasibility. The deceleration constraint holds by construction. For the acceleration constraint at i: if $w_i^b = w_i^f$ then $w_{i+1}^b \leq w_{i+1}^f \leq w_i^f + 2a_i\Delta_i = w_i^b + 2a_i\Delta_i$, and otherwise $w_{i+1}^b - w_i^b = -2d_i\Delta_i < 2a_i\Delta_i$. *Maximality.* A feasible w satisfies $w \leq w^f$ by forward induction, and then $w \leq w^b$ by backward induction. *Time.* Constant acceleration gives $\Delta_i = \frac{1}{2}(v_i + v_{i+1})t$. □

The implementation evaluates a velocity-dependent bound at the start of each segment. Since the voltage bound on acceleration falls with speed, this is optimistic by $O(\Delta)$, with samples about 1 cm apart. The generator therefore uses only $U = 10$ V of the available 12 V, and this margin also leaves room for feedback.

5.3. **Turns in place.** For a turn through D under $|\dot{\theta}| \leq \Omega$, $|\ddot{\theta}| \leq \Lambda$, Pontryagin's principle gives the accelerate, coast, brake profile, of duration $D/\Omega + \Omega/\Lambda$ when $D \geq \Omega^2/\Lambda$ and $2\sqrt{D/\Lambda}$ otherwise. A turn is tracked by feedforward from the profile plus proportional-derivative feedback with gains (k_P, k_D). If the wheels realize the commanded rate exactly, the error obeys $(1 + k_D)\dot{e} = -k_P e$, a first-order decay with time constant $(1 + k_D)/k_P$, or 0.21 s for the default gains.

6. Tracking laws

6.1. **Error coordinates.** We use two sets of coordinates, shown in Figure 3. Let the reference pose be (p_r, θ_r) with speeds $(v_\mathrm{d}, \omega_\mathrm{d})$, $\dot{p}_r = v_\mathrm{d}\mathbf{e}(\theta_r)$, $\dot{\theta}_r = \omega_\mathrm{d}$. In the *robot frame*,

$$\begin{pmatrix} e_x \\ e_y \end{pmatrix} = \mathrm{R}(-\theta)(p_r - p), \qquad e_\theta = \theta_r - \theta. \tag{6.1}$$

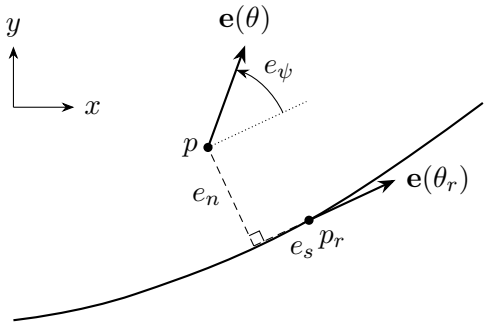


FIGURE 3. Reference-frame error coordinates. The body-frame coordinates (e_x, e_y) are the same vector, negated and resolved along the robot's axes.

Differentiating and using (2.3) gives

$$\dot{e}_x = \omega e_y - v + v_{\mathrm{d}} \cos e_\theta, \qquad \dot{e}_y = -\omega e_x + v_{\mathrm{d}} \sin e_\theta, \qquad \dot{e}_\theta = \omega_{\mathrm{d}} - \omega. \tag{6.2}$$

In the *reference frame*, used for prediction,

$$\begin{pmatrix} e_s \\ e_n \end{pmatrix} = \mathrm{R}(-\theta_r)(p - p_r), \qquad e_\psi = \theta - \theta_r, \tag{6.3}$$

with

$$\dot{e}_s = \omega_{\mathrm{d}} e_n + v \cos e_\psi - v_{\mathrm{d}}, \qquad \dot{e}_n = -\omega_{\mathrm{d}} e_s + v \sin e_\psi, \qquad \dot{e}_\psi = \omega - \omega_{\mathrm{d}}. \tag{6.4}$$

The two are related by $(e_x, e_y)^{\mathsf{T}} = -\mathrm{R}(e_\theta)(e_s, e_n)^{\mathsf{T}}$ and $e_\theta = -e_\psi$.

6.2. **RAMSETE and its strict Lyapunov function.** The law is

$$v^\star = v_{\mathrm{d}} \cos e_\theta + k e_x, \qquad \omega^\star = \omega_{\mathrm{d}} + k e_\theta + b\, v_{\mathrm{d}}\, \mathrm{sinc}(e_\theta)\, e_y, \qquad k = 2\zeta \sqrt{\omega_{\mathrm{d}}^2 + b v_{\mathrm{d}}^2}, \tag{6.5}$$

with $b > 0$ and $0 < \zeta \le 1$. Let $V = e_x^2 + e_y^2 + e_\theta^2/b$.

Proposition 6.1. *With $v = v^\star + \delta v$ and $\omega = \omega^\star + \delta\omega$ in (6.2),*

$$\tfrac{1}{2}\dot{V} = -k(e_x^2 + e_\theta^2/b) - e_x \delta v - \frac{e_\theta}{b}\delta\omega. \tag{6.6}$$

Proof. From (6.2), $\frac{1}{2}\dot{V} = e_x(v_{\mathrm{d}} \cos e_\theta - v) + v_{\mathrm{d}} e_y \sin e_\theta + (e_\theta/b)(\omega_{\mathrm{d}} - \omega)$, because the terms in $\omega e_x e_y$ cancel. Substituting (6.5), the two terms in $v_{\mathrm{d}} e_y \sin e_\theta$ cancel, since $e_\theta \,\mathrm{sinc}\, e_\theta = \sin e_\theta$. □

At $\delta v = \delta\omega = 0$ the right side of (6.6) vanishes on the entire axis $e_x = e_\theta = 0$. So V certifies stability, and convergence needs an additional invariance argument, which gives no rate. It also cannot be used as a constraint on another controller: a plan that ignores a pure cross-track error satisfies $\dot{V} \le 0$ forever. We therefore make the function strict.

Definition 6.2. For $\varepsilon \ge 0$ and $v_{\mathrm{d}} > 0$, let

$$W_\varepsilon = e_x^2 + e_y^2 + \frac{e_\theta^2}{b} + 2\varepsilon\, e_y \sin e_\theta. \tag{6.7}$$

The case $v_{\mathrm{d}} < 0$ reduces to this one. The substitution $\tilde{\theta} = \theta + \pi$, $\tilde{v} = -v$ maps (6.5) to itself with (e_x, e_y) negated and v_{d} positive, and W_ε then acquires a minus sign on the cross term. The implementation uses $\mathrm{sgn}(v_{\mathrm{d}})\,\varepsilon$. From here on we state results for $v_{\mathrm{d}} > 0$.

Let $\kappa = \omega_{\mathrm{d}}/v_{\mathrm{d}}$ be the reference curvature and $K = k/v_{\mathrm{d}} = 2\zeta\sqrt{\kappa^2 + b}$. Define

$$\Phi_\varepsilon(e;\kappa) = -Ke_x^2 - \frac{K}{b}e_\theta^2 - \varepsilon b\,\mathrm{sinc}(e_\theta)\cos(e_\theta)\,e_y^2 + \varepsilon\sin^2 e_\theta \\ - \varepsilon K e_y e_\theta \cos e_\theta - \varepsilon\big(\kappa + Ke_\theta + b\,\mathrm{sinc}(e_\theta)e_y\big)e_x \sin e_\theta. \quad (6.8)$$

Lemma 6.3 (Derivative)**.** *With $v = v^\star + \delta v$, $\omega = \omega^\star + \delta\omega$,*

$$\tfrac{1}{2}\dot{W}_\varepsilon = v_{\mathrm{d}}\,\Phi_\varepsilon(e;\kappa) - g_v\,\delta v - g_\omega\,\delta\omega, \qquad g_v = e_x, \quad g_\omega = \frac{e_\theta}{b} + \varepsilon(e_x\sin e_\theta + e_y\cos e_\theta). \quad (6.9)$$

In particular, under (6.5) *the derivative of W_ε with respect to the reference arc length s, $\mathrm{d}s = v_{\mathrm{d}}\,\mathrm{d}t$, is $2\Phi_\varepsilon$, and it does not depend on the speed.*

Proof. Write $W_\varepsilon = V + 2\varepsilon e_y \sin e_\theta$. Then $\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}(2\varepsilon e_y \sin e_\theta) = \varepsilon(\dot{e}_y \sin e_\theta + e_y \cos e_\theta\,\dot{e}_\theta)$. Insert (6.2). The dependence on ω is $-\varepsilon\omega(e_x \sin e_\theta + e_y \cos e_\theta)$, which with (6.6) gives g_ω, and there is no further dependence on v. At $\delta v = \delta\omega = 0$, every term has a factor of v_{d} after using $k = Kv_{\mathrm{d}}$ and $\omega^\star = v_{\mathrm{d}}(\kappa + Ke_\theta + b\,\mathrm{sinc}(e_\theta)e_y)$; collecting them gives (6.8). □

Near the origin Φ_ε is the quadratic form $-\zeta^{\mathsf{T}} M\zeta$ in $\zeta = (e_x, e_\theta, e_y)$, with

$$M(\kappa,\varepsilon) = \begin{pmatrix} K & \varepsilon\kappa/2 & 0 \\ \varepsilon\kappa/2 & K/b - \varepsilon & \varepsilon K/2 \\ 0 & \varepsilon K/2 & \varepsilon b \end{pmatrix}, \quad (6.10)$$

and likewise $W_\varepsilon = \zeta^{\mathsf{T}} D_\varepsilon \zeta + O(\|\zeta\|^4)$ with $D_\varepsilon = \left(\begin{smallmatrix} 1 & 0 & 0 \\ 0 & 1/b & \varepsilon \\ 0 & \varepsilon & 1 \end{smallmatrix}\right)$.

Theorem 6.4 (Strict Lyapunov function)**.** *Let $b > 0$, $0 < \zeta \le 1$ and $\varepsilon > 0$.*

(1) *If $\varepsilon\sqrt{b} < 1$, then with $N = e_x^2 + e_y^2 + e_\theta^2/b$,*

$$(1 - \varepsilon\sqrt{b})N \le W_\varepsilon \le (1 + \varepsilon\sqrt{b})N. \quad (6.11)$$

(2) *$M(\kappa,\varepsilon) \succ 0$ if and only if*

$$K^2 - \varepsilon\big(bK + \tfrac{1}{4}K^3\big) - \tfrac{1}{4}\varepsilon^2\kappa^2 b > 0, \quad (6.12)$$

that is, if and only if $\varepsilon < \varepsilon^\star(\kappa)$, where $\varepsilon^\star$ is the positive root of the left side. At $\kappa = 0$ it equals $2\zeta/(\sqrt{b}(1 + \zeta^2))$.

(3) *Let $\lambda(\kappa,\varepsilon) > 0$ be the smallest generalized eigenvalue of (M, D_ε). For every $\lambda' < \lambda$ there is a neighborhood of the origin on which $\Phi_\varepsilon \le -\lambda' W_\varepsilon$. Hence along* (6.5)*, for a reference of constant curvature, $W_\varepsilon(s) \le W_\varepsilon(0)e^{-2\lambda' s}$ near the origin. The same holds for time-varying curvature with $\inf_{|\kappa|\le\bar{\kappa}}\lambda$ in place of λ.*

(4) *Let $A > 0$, $0 < \delta < \pi/2$ and $\sigma_\delta = \sin(2\delta)/(2\delta)$. If $|e_x| \le A$, $|e_\theta| \le \delta$, and*

$$c_0 := \frac{K}{b} - \varepsilon\Big(1 + KA + \frac{(K + bA)^2}{2b\sigma_\delta}\Big) - \frac{\varepsilon^2\kappa^2}{2K} > 0, \quad (6.13)$$

then

$$-\Phi_\varepsilon \ge \frac{K}{2}e_x^2 + \frac{\varepsilon b\sigma_\delta}{2}e_y^2 + c_0 e_\theta^2. \quad (6.14)$$

Thus for every compact set of errors with $|e_\theta| < \pi/2$ there is an $\varepsilon > 0$ such that W_ε decreases exponentially in s on a sublevel set containing it.

Proof. (1) This follows from $|2\varepsilon e_y \sin e_\theta| \le 2\varepsilon|e_y||e_\theta| \le \varepsilon\sqrt{b}(e_y^2 + e_\theta^2/b)$.

(2) The leading principal minors of M are K, $m_2 = K(K/b - \varepsilon) - \varepsilon^2\kappa^2/4$, and

$$\det M = \varepsilon b\, m_2 - \tfrac{1}{4}\varepsilon^2 K^3 = \varepsilon\big[K^2 - \varepsilon bK - \tfrac{1}{4}\varepsilon K^3 - \tfrac{1}{4}\varepsilon^2\kappa^2 b\big].$$

If $\det M > 0$ then $m_2 > \varepsilon K^3/(4b) > 0$, so the sign of the determinant alone decides. As a function of $\varepsilon \ge 0$ the bracket is positive at 0 and strictly decreasing, so it has exactly one positive root. At $\kappa = 0$, $K = 2\zeta\sqrt{b}$ and the condition reads $\varepsilon < K/(b + K^2/4) = 2\zeta/(\sqrt{b}(1+\zeta^2))$.

(3) We have $\zeta^{\mathsf{T}} M\zeta \ge \lambda\zeta^{\mathsf{T}} D_\varepsilon \zeta$ by definition, and Φ_ε, W_ε differ from their quadratic parts by $O(\|\zeta\|^3)$. Apply Lemma 6.3, and use the fact that W_ε does not depend on the reference, so that its sublevel sets are invariant.

(4) Write $a = |e_x|$, $c = |e_\theta|$, $d = |e_y|$. In (6.8) we use $\operatorname{sinc}(e_\theta)\cos(e_\theta) = \sin(2e_\theta)/(2e_\theta) \ge \sigma_\delta$, $\sin^2 e_\theta \le c^2$, $|\sin e_\theta| \le c$, and $|\operatorname{sinc}| \le 1$. Then

$$-\Phi_\varepsilon \ge Ka^2 + \big(\tfrac{K}{b} - \varepsilon - \varepsilon KA\big)c^2 + \varepsilon b\sigma_\delta d^2 - \varepsilon(K + bA)cd - \varepsilon|\kappa|ac.$$

By Young's inequality, $\varepsilon|\kappa|ac \le \frac{K}{2}a^2 + \frac{\varepsilon^2\kappa^2}{2K}c^2$ and $\varepsilon(K+bA)cd \le \frac{\varepsilon b\sigma_\delta}{2}d^2 + \frac{\varepsilon(K+bA)^2}{2b\sigma_\delta}c^2$. This gives (6.14). Since K is bounded above and below on $|\kappa| \le \bar\kappa$, condition (6.13) holds for small ε. Then $-\Phi_\varepsilon \ge \mu N \ge \mu W_\varepsilon/(1+\varepsilon\sqrt{b})$ by (6.11), with $\mu = \min(K/2, \varepsilon b\sigma_\delta/2, bc_0)$. A sublevel set $\{W_\varepsilon \le c\}$ with $c/(1-\varepsilon\sqrt{b}) \le \min(A^2, b\delta^2)$ lies in the box and is invariant. □

Remark 6.5 (Sharpness). The linearization of (6.5) on a straight line, in arc length, has the (e_y, e_θ) block $\left(\begin{smallmatrix} 0 & 1 \\ -b & -K \end{smallmatrix}\right)$, with eigenvalues $-\zeta\sqrt{b} \pm i\sqrt{b(1-\zeta^2)}$. So for $\zeta < 1$ no certificate can show a rate above $\zeta\sqrt{b}$. For $b = 15$ and $\zeta = 0.7$ this is 2.7111 per meter, and maximizing $\lambda(0,\varepsilon)$ over ε gives 2.7105, at $\varepsilon = 0.181$. The certificate is nearly tight. Figure 4 shows V and W_ε along a trajectory.

Remark 6.6 (Values used). For $b = 15$, $\zeta = 0.7$, the threshold $\varepsilon^\star(\kappa)$ lies between 0.238 and 0.246 for $|\kappa| \le 5\ \mathrm{m}^{-1}$, and falls below 0.1 only at $|\kappa| = 26.6\ \mathrm{m}^{-1}$, a radius of under 4 cm. Since $\varepsilon\sqrt{b} < 1$ needs $\varepsilon < 0.258$, both requirements are met by the value $\varepsilon = 0.1$ in the implementation, with $\lambda(0, 0.1) = 1.397$. Condition (6.13) is far from sharp, however: for $\varepsilon = 0.1$ it fails on the box $A = 0.3$, $\delta = 0.6$ that the controller of Section 7 needs. For the implemented gains we therefore bound Φ_ε directly.

Proposition 6.7 (Computer-assisted bound)**.** *Let $b = 15$, $\zeta = 0.7$, $\varepsilon = 0.1$ and $|\kappa| \le 4\ m^{-1}$. Then for every e_x,*

$$-\Phi_\varepsilon \ge 0.95\, W_\varepsilon \qquad \text{on } \{|e_\theta| \le 0.6,\ |e_y| \le 0.3\},$$
$$-\Phi_\varepsilon \ge 0.35\, W_\varepsilon \qquad \text{on } \{|e_\theta| \le 1.2,\ |e_y| \le 0.6\}.$$

The sublevel sets $\{W_\varepsilon \le 0.0208\}$ and $\{W_\varepsilon \le 0.0873\}$ lie in these two regions, and they are invariant under (6.5) *for references with $|\kappa| \le 4\ m^{-1}$.*

Proof. Put $S = \operatorname{sinc} e_\theta$ and $C = \cos e_\theta$, so that $\sin e_\theta = Se_\theta$. From (6.7) and (6.8), $-\Phi_\varepsilon - \lambda W_\varepsilon = \xi^{\mathsf{T}} Q\xi$ with $\xi = (e_x, e_\theta, e_y)$ and Q symmetric, $Q_{13} = 0$, and

$$Q_{11} = K - \lambda, \quad Q_{22} = \frac{K}{b} - \varepsilon S^2 - \frac{\lambda}{b}, \quad Q_{33} = \varepsilon bSC - \lambda,$$
$$Q_{12} = \frac{\varepsilon}{2}(\kappa + Ke_\theta + bSe_y)S, \quad Q_{23} = \frac{\varepsilon KC}{2} - \lambda\varepsilon S.$$

The matrix depends only on (e_θ, e_y, κ), so e_x is unrestricted. On a box of those three variables, interval arithmetic gives symmetric matrices Q_c and Q_r with $|Q - Q_c| \le Q_r$ entrywise. By Weyl's inequality, $\lambda_{\min}(Q) \ge \lambda_{\min}(Q_c) - \|Q_r\|_F$, which is a simple sufficient case of the conditions

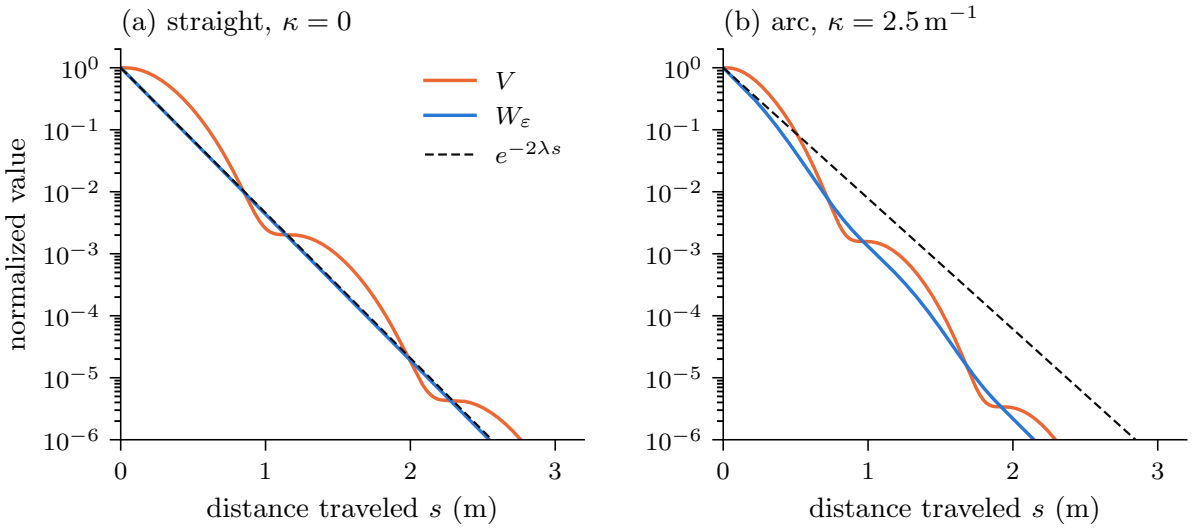


FIGURE 4. Decay against distance for an initial cross-track error of 10 cm under the kinematic closed loop (6.5), with $b = 15$, $\zeta = 0.7$ and $\varepsilon = 0.18$. The classical function V has plateaus where e_x and e_θ cross zero. The strict function W_ε does not, and it lies on the certified envelope $e^{-2\lambda s}$ in (a) and below it in (b).

for interval matrices studied by Rohn [36]. Adaptive bisection reaches a positive value on every box, with 90,199 boxes for the first claim and 132,215 for the second. The computation uses double precision with a margin of 10^{-9} on the eigenvalue test and 10^{-12} on each radius, in place of directed rounding.

For the sublevel sets: minimizing (6.7) over e_y at fixed e_θ gives $e_\theta^2/b - \varepsilon^2 \sin^2 e_\theta$, which increases with $|e_\theta|$ and equals 0.0208 at 0.6 and 0.0873 at 1.2. The corresponding limits from e_y are larger. Invariance for $v_\mathrm{d} > 0$ follows from Lemma 6.3 and the two inequalities, since W_ε does not depend on the reference. □

A numerical minimization of $-\Phi_\varepsilon / W_\varepsilon$ on the first region gives about 1.25, so the first constant is not tight.

6.3. **The linear quadratic regulator.** Linearizing (6.2) about the reference on a straight line, with $\omega_\mathrm{d} = 0$, gives a system in $e = (e_x, e_y, e_\theta)$ and speed corrections $(\delta v, \delta\omega)$ with

$$A = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & v_\mathrm{d} \\ 0 & 0 & 0 \end{pmatrix}, \qquad B = \begin{pmatrix} -1 & 0 \\ 0 & 0 \\ 0 & -1 \end{pmatrix}.$$

Here $A^2 = 0$, so the exact discretization over a step h is $A_d = I + hA$, $B_d = (hI + \frac{h^2}{2}A)B$, and the pair is controllable exactly when $v_\mathrm{d} \neq 0$. A table of gains over speed, from the discrete Riccati equation, gives the LTV-LQR tracker used as a comparison below. The essential behavior can be read from the steering pair alone.

Proposition 6.8. *Let $\dot{x}_1 = v x_2$, $\dot{x}_2 = u$ with $v > 0$, cost $\int (q_1 x_1^2 + q_2 x_2^2 + r u^2)\,\mathrm{d}t$. The algebraic Riccati equation has the solution*

$$P = \begin{pmatrix} p_1 & p_2 \\ p_2 & p_3 \end{pmatrix}, \quad p_2 = \sqrt{q_1 r}, \quad p_3 = \sqrt{r(q_2 + 2 v p_2)}, \quad p_1 = \frac{p_2 p_3}{r v}, \tag{6.15}$$

with gain $(\sqrt{q_1/r},\ p_3/r)$*. In particular the gain on* x_1 *does not depend on* v*, while the cost-to-go of the cross-track state satisfies* $p_1 \sim \sqrt{q_1 q_2}/v$ *as* $v \to 0$*.*

Proof. The entries of $A^{\mathsf T}P + PA - PBr^{-1}B^{\mathsf T}P + Q = 0$ read $q_1 - p_2^2/r = 0$, then $vp_1 - p_2p_3/r = 0$, and then $2vp_2 + q_2 - p_3^2/r = 0$. □

The blow-up as $v \to 0$ is the Riccati form of the fact that a stopped robot cannot correct a cross-track error, and it will decide how the terminal cost of Section 7.3 behaves near a stop.

6.4. **The pose stabilizer.** Point-to-point motions use polar coordinates about the goal (p_g, θ_g): with ϕ the bearing of the goal from the robot, put $\rho = \|p_g - p\|$, $\alpha = \phi - \theta$, $\beta = \phi - \theta_g$. Then

$$\dot\rho = -v\cos\alpha, \qquad \dot\alpha = \frac{v\sin\alpha}{\rho} - \omega, \qquad \dot\beta = \frac{v\sin\alpha}{\rho}.$$

Proposition 6.9 (Aicardi et al. [1])**.** *With gains* $\gamma, k, h > 0$*, the law*

$$v = \gamma\rho\cos\alpha, \qquad \omega = k\alpha + \gamma\frac{\cos\alpha\sin\alpha}{\alpha}(\alpha + h\beta) \tag{6.16}$$

gives, for $U = \tfrac12(\rho^2 + \alpha^2 + h\beta^2)$*,*

$$\dot U = -\gamma\rho^2\cos^2\alpha - k\alpha^2 \le 0,$$

and every trajectory converges to $(\rho, \alpha, \beta) = (0, 0, 0)$*.*

Proof. The closed loop is smooth in (ρ, α, β): $\dot\rho = -\gamma\rho\cos^2\alpha$, $\dot\alpha = -k\alpha - \gamma(\cos\alpha\sin\alpha/\alpha)h\beta$, $\dot\beta = \gamma\cos\alpha\sin\alpha$. The derivative of U follows by substitution, the terms in β cancelling. On the largest invariant set in $\{\dot U = 0\}$ we have $\alpha \equiv 0$, so $\rho = 0$; and $\dot\alpha = 0$ forces $\beta = 0$. LaSalle's principle applies on the compact sublevel sets of U. □

Discontinuity in Cartesian coordinates at the goal is what evades Brockett's obstruction [9]. Linearized at the goal, $\ddot\beta + k\dot\beta + \gamma^2 h\beta = 0$ and $\dot\rho = -\gamma\rho$. For the default gains $\gamma = 2$, $k = 5$, $h = 1$, the roots are -1 and -4.

The same law has a practical drawback, which is why it is not used to finish a path. If the robot arrives with a small pure sideways miss, then α is near $\pm\pi/2$ however small ρ is, and (6.16) turns the robot toward the goal. The motion converges, but with a large excursion of heading for a miss of a centimeter.

7. Predictive tracking

7.1. **Prediction model.** The state is $x = (e_s, e_n, e_\psi, \xi_L, \xi_R)^{\mathsf T}$, with ξ_i the wheel speed error against the reference. Linearizing (6.4) about $v = v_{\mathrm d}$ gives $\dot e_s = \omega_{\mathrm d} e_n + \delta v$, $\dot e_n = -\omega_{\mathrm d} e_s + v_{\mathrm d} e_\psi$, $\dot e_\psi = \delta\omega$, where by (2.4) $\delta v = \tfrac12(\xi_L + \xi_R)$ and $\delta\omega = (\xi_R - \xi_L)/\ell$. Writing $E = (e_s, e_n)^{\mathsf T}$ and $f = (\delta v, v_{\mathrm d} e_\psi)^{\mathsf T}$, and letting φ be the angle through which the reference turns,

$$E(h) = \mathrm R(-\varphi(h))\Big(E(0) + \int_0^h \mathrm R(\varphi(t))f(t)\,\mathrm dt\Big). \tag{7.1}$$

We approximate $\mathrm R(\varphi(t)) \approx I$ inside the integral and use the wheel averages of Lemma 3.3. For node k of length h_k on which the reference turns through φ_k at mean speed $\bar v_k$, with decisions δ_L, δ_R held constant, this gives

$$\begin{aligned} \bar\xi_i &= a_1\xi_i + a_2\delta_i, \qquad \delta\bar v = \tfrac12(\bar\xi_L + \bar\xi_R), \qquad \delta\bar\omega = (\bar\xi_R - \bar\xi_L)/\ell, \\ \begin{pmatrix} e_s^+ \\ e_n^+ \end{pmatrix} &= \mathrm R(-\varphi_k)\begin{pmatrix} e_s + h_k\delta\bar v \\ e_n + h_k\bar v_k(e_\psi + \tfrac12 h_k\delta\bar\omega) \end{pmatrix}, \\ e_\psi^+ &= e_\psi + h_k\delta\bar\omega, \qquad \xi_i^+ = \alpha\xi_i + \gamma\delta_i. \end{aligned} \tag{7.2}$$

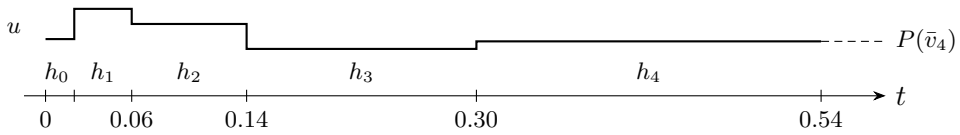


FIGURE 5. The prediction grid. Decisions are held on each node, and the Riccati cost P stands in for all later time.

Thus $x_{k+1} = A_k x_k + B_k u_k$ with $u_k = (\delta_L, \delta_R)_k$, where A_k and B_k depend on the reference only through φ_k and $\bar{v}_k$. The wheel coefficients of Lemma 3.3 depend only on h_k and are precomputed.

The plan has $N = 5$ nodes of lengths $0.02, 0.04, 0.08, 0.16, 0.24$ s, 0.54 s in all (Figure 5). The reference at each node boundary is sampled from the trajectory, using φ_k as the exact heading difference. Each wheel's mean speed $\bar{v}_{i,k}$ and mean acceleration $(v_{i,k+1} - v_{i,k})/h_k$ give a feedforward

$$u^{\mathrm{ff}}_{i,k} = k_S \operatorname{sgn} \bar{v}_{i,k} + k_V \bar{v}_{i,k} + k_A \frac{v_{i,k+1} - v_{i,k}}{h_k} + \hat{\eta}_i \min(|\bar{v}_{i,k}|/v_\ell, 1), \tag{7.3}$$

where $\hat{\eta}_i$ is the learned voltage of Section 7.8. Taking finite differences of wheel speeds captures the rate of change of curvature, which the product $a\kappa$ alone would miss, and it makes the mean acceleration over a node exact for a piecewise-linear speed.

7.2. **The quadratic program.** Let $Q = \operatorname{diag}(q_s, q_n, q_\psi, q_w, q_w)$ and $q_u > 0$. With $z \in \mathbb{R}^{2N}$ collecting the decisions, the cost is the discretized integral

$$J(z) = \sum_{k=0}^{N-1} h_k (x_{k+1}^{\mathsf{T}} Q x_{k+1} + q_u \|u_k\|^2) + x_N^{\mathsf{T}} P\, x_N, \tag{7.4}$$

to be minimized over the box

$$-U - u^{\mathrm{ff}}_{i,k} \le \delta_{i,k} \le U - u^{\mathrm{ff}}_{i,k}. \tag{7.5}$$

The weights follow Bryson's rule, each the inverse square of an acceptable value (Appendix B). Writing $x_k = \Phi_k x_0 + \Gamma_k z$ for the rollout, with $\Phi_{k+1} = A_k \Phi_k$ and $\Gamma_{k+1} = A_k \Gamma_k + [\,0\ B_k\ 0\,]$,

$$\begin{gathered} J(z) = z^{\mathsf{T}} H z + 2 g^{\mathsf{T}} z + \text{const}, \\ H = \sum_{k=1}^{N} h_{k-1} \Gamma_k^{\mathsf{T}} Q \Gamma_k + \Gamma_N^{\mathsf{T}} P \Gamma_N + q_u D_h, \qquad g = \Big(\sum_{k=1}^{N} h_{k-1} \Gamma_k^{\mathsf{T}} Q \Phi_k + \Gamma_N^{\mathsf{T}} P \Phi_N \Big) x_0, \end{gathered} \tag{7.6}$$

where D_h is the diagonal matrix of the h_k, each repeated twice.

Lemma 7.1. *If $P \succeq 0$ then $H \succeq q_u h_{\min} I$. Hence (7.4)–(7.5) has a unique minimizer for every state. The box is nonempty for all x_0, since each interval has width $2U$.*

Proof. The sum in H is positive semidefinite and $q_u D_h \succeq q_u h_{\min} I$. A strictly convex quadratic attains a unique minimum on a nonempty compact convex set. □

Since voltage limits do not depend on the state, the usual difficulties of recursive feasibility do not arise, and the controller needs no slack variables.

7.3. **The terminal cost.** The matrix P is taken from a table of discrete Riccati solutions for the model (7.2) with $\varphi = 0$ at a step $h_T = 0.05$ s, with weights $h_T Q$ and $h_T q_u I$. Let $A(v), B(v)$ be these matrices, and $P(v)$ the stabilizing solution.

Proposition 7.2 (Mirror symmetry)**.** *Let* $T = \operatorname{diag}(1, -1, 1, 1, 1)$. *For* $v \neq 0$, $P(-v) = TP(v)T$.

Proof. In (7.2) with $\varphi = 0$, the speed $\bar{v}$ appears only in the equation for e_n. Reversing the signs of e_n and of v together leaves the equations unchanged, which says that $A(-v) = TA(v)T$ and $B(-v) = TB(v)$, and also $TQT = Q$. Substituting into the Riccati equation shows that $TP(v)T$ solves it for $-v$. The pair $(A(v), B(v))$ is stabilizable for $v \neq 0$ and $Q \succ 0$, so the stabilizing solution is unique. □

So the table covers $v \geq 0$ only, at eight speeds, and the implementation stores 200 numbers where 375 would otherwise be needed. Proposition 6.8 says that $P_{nn}(v)$ diverges as $v \to 0$. The computed values show this: $P_{nn} \approx 10290,\ 5289,\ 2787,\ 1534,\ 904$ at $v = 0.05, 0.1, 0.2, 0.4, 0.8$ m/s. Near the end of a path, this would produce large turning commands to correct a cross-track error that can no longer be corrected. The table's first entry is therefore computed with $q_n = 0$ at $v = 0$:

Proposition 7.3. *With* $v = 0$ *and* $q_n = 0$, *the Riccati iteration from* $P_0 = Q$ *converges to a limit whose row and column for* e_n *vanish.*

Proof. At $v = 0$ and $\varphi = 0$, column n of A is the unit vector $\mathbf{e}_n$, and row n of B and the off-diagonal part of row n of A vanish. So if $P_j \mathbf{e}_n = 0$ then $A^{\mathsf{T}} P_j A \mathbf{e}_n = A^{\mathsf{T}} P_j \mathbf{e}_n = 0$, and $B^{\mathsf{T}} P_j A \mathbf{e}_n = 0$, and hence $P_{j+1} \mathbf{e}_n = Q \mathbf{e}_n = 0$. The remaining four states form a stabilizable and detectable system. □

Values between table entries are found by linear interpolation in $|v|$. A convex combination of positive semidefinite matrices is positive semidefinite, so Lemma 7.1 applies, and the weight on cross-track error fades continuously to zero at a stop. This fade is what prevents a spin at the end of a path.

7.4. **An exact solver.** We solve the problem

$$\min_{z \in \mathbb{R}^n} \tfrac{1}{2} z^{\mathsf{T}} H z + g^{\mathsf{T}} z \quad \text{subject to} \quad \underline{z} \leq z \leq \overline{z}, \tag{7.7}$$

with $H \succ 0$. Each variable carries a state in $\{\mathsf{F}, \mathsf{L}, \mathsf{U}\}$: free, at the lower bound, or at the upper bound. Given states, the *subproblem* fixes bound variables at their bounds and solves $H_{FF} z_F = -(g_F + H_{FB} z_B)$ by Cholesky factorization. Let $G = Hz + g$. A variable is *wrong* if it is free and outside its bounds, or if it is at its lower bound with $G_i < 0$, or at its upper bound with $G_i > 0$. Algorithm 1 states the method. It moves all wrong variables at once while their number keeps falling, and it reverts to moving only the wrong variable of highest index when that number has failed to fall for n_p iterations.

Theorem 7.4. *Let* $H \succ 0$ *and* $\underline{z} < \overline{z}$, *in exact arithmetic.*

1. *If no variable is wrong,* z *is the unique solution of* (7.7).
2. *From any initial states, the rule that always moves only the wrong variable of highest index stops after at most* $3^n - 1$ *moves.*
3. *Algorithm 1 with* $k_{\max}$ *unlimited stops after at most* $(n(n_p + 1) + 1)\, 3^n$ *iterations, with the exact solution.*
4. *If stopped early, it returns a point of the box.*

Algorithm 1 Box QP by block principal pivoting with a least-index fallback

Require: states from the previous solve (warm start); iteration limit $k_{\max}$; patience $n_p \geq 0$
$b \leftarrow n+1$; $c \leftarrow n_p$
for $i = 1, \ldots, k_{\max}$ **do**
  solve the subproblem for z; let $\mathcal{W}$ be the set of wrong variables
  if $\mathcal{W} = \emptyset$ **then return** z
  end if
  if $|\mathcal{W}| < b$ **then** $b \leftarrow |\mathcal{W}|$; $c \leftarrow n_p$
  else if $c > 0$ **then** $c \leftarrow c - 1$
  else $\mathcal{W} \leftarrow \{\max \mathcal{W}\}$
  end if
  for $j \in \mathcal{W}$ **do**
    if j is free, set it to the bound it violates; otherwise set it free
  end for
end for
return $\mathrm{clip}(z, \underline{z}, \overline{z})$

Proof. (1) The Karush–Kuhn–Tucker conditions, which are sufficient for a convex problem, say that for each i either $z_i = \underline{z}_i$ and $G_i \geq 0$, or $z_i = \overline{z}_i$ and $G_i \leq 0$, or $\underline{z}_i < z_i < \overline{z}_i$ and $G_i = 0$. Absence of wrong variables gives exactly this; a free variable has $G_i = 0$ by the subproblem. Uniqueness is strict convexity.

(2) We use induction on n. Let f be the objective.

For $n = 1$, let $t^\star$ be the minimizer of f on $\mathbb{R}$. If $t^\star \in [\underline{z}, \overline{z}]$, then F is not wrong and L, U each reach F in at most one move. If $t^\star < \underline{z}$, then L is not wrong, F moves to L, and U moves to F, giving at most two moves. The case $t^\star > \overline{z}$ is symmetric. So the number of moves is at most $2 = 3^1 - 1$.

Let $n > 1$, and write $z = (t, z')$ with $t = z_1$ the variable of lowest priority, which moves only when it is the sole wrong variable. While the state of t does not change, the remaining variables follow the rule for an $(n-1)$-dimensional problem of the same form: if t is bound, it is $f(\underline{z}_1, \cdot)$ or $f(\overline{z}_1, \cdot)$; if t is free, it is the partial minimum $z' \mapsto \min_{t \in \mathbb{R}} f(t, z')$, a strictly convex quadratic (the Schur complement of H), and the free solve for $(t, z_{F'})$ coincides with the solve for $z_{F'}$ in the reduced problem, with the same gradients on z' because $\partial_t f = 0$ there. By induction, at most $3^{n-1} - 1$ moves occur before z' has no wrong variable. Then z' solves the corresponding box problem.

Let $\varphi(t) = \min_{z' \in \mathcal{B}'} f(t, z')$, with $\mathcal{B}'$ the box in z'. It is strictly convex on $\mathbb{R}$, coercive, and C^1 with $\varphi'(t) = \partial_t f(t, z'(t))$, since the inner minimizer $z'(t)$ is unique and continuous. Let $t^\star$ minimize φ. When z' is solved: in state L, t is wrong if and only if $\varphi'(\underline{z}_1) < 0$, that is, $\underline{z}_1 < t^\star$; in state U, if and only if $\overline{z}_1 > t^\star$; and in state F, the solved point has $\partial_t f = 0$ with z' optimal for the reduced problem, so $t = t^\star$, and t is wrong if and only if $t^\star \notin [\underline{z}_1, \overline{z}_1]$. This is the same transition structure as in the case $n = 1$, so t moves at most twice. In all there are at most three rounds of at most $3^{n-1} - 1$ moves, plus two moves of t, which is $3^n - 1$.

(3) The test $|\mathcal{W}| < b$ succeeds at most n times, because b strictly decreases through values in $\{1, \ldots, n\}$. Each success allows at most n_p further block moves. So there are at most $B = n(n_p + 1)$ block moves. They divide the remaining iterations into at most $B + 1$ runs of single moves, and each run has at most $3^n - 1$ moves, because (2) holds from any initial states. With the final iteration, which finds no wrong variable, the total is at most $B + (B+1)(3^n - 1) + 1 = (B+1)3^n$.

(4) The last line clips. □

Remark 7.5. The bound 3^n is a worst case. In the experiments of Section 9 the solver averaged 1.01 iterations per control cycle and never took more than three, on problems with $n = 10$, because the warm start is nearly right. Random problems with $n = 10$ and condition number 10^6 took a mean of 7.3 and a maximum of 75 iterations from a cold start. The implementation sets $k_{\max} = 6$, so the worst-case time is fixed in advance, and it compares against a tolerance of 10^{-9} in floating point, where the theorem does not strictly apply.

7.5. **The plan clock.** A reference indexed by time runs away from a robot that cannot keep up, and a tracking law then cuts corners to catch it. We separate geometry from timing, as in on-line time scaling [19, 13]. Let the reference $q(\tau) = (p_r, \theta_r, v_r, \kappa)(\tau)$ be evaluated at a *plan clock* $\tau(t)$ with

$$\dot\tau = r(t) \in [0, 1]. \tag{7.8}$$

Proposition 7.6. *The curve $p_r(\tau(t))$ has the same image as p_r and the same curvature at each point. The scaled reference has speed rv_r, turn rate $r\omega_r$, and acceleration $r^2 a_r + \dot r v_r$.*

Proof. Apply the chain rule, and note that the curvature is a function of the arc length and not of its parameter. □

The prediction model uses rv_r and $r\omega_r$, and it samples the reference at $\tau + r\sum_{j<k} h_j$, treating r as constant over the plan. The target rate is the smaller of two limits.

A voltage limit. Let p be the largest magnitude of (7.3) over all nodes and wheels at the current rate, and $\bar U = 0.95\,U$ the usable voltage, the rest being left for feedback. Then

$$r_u = \max(r, r_0)\,\frac{\bar U - k_S}{\max(p - k_S,\ 10^{-3})}, \tag{7.9}$$

with $r_0 = 0.05$. The floor r_0 is needed because otherwise $r = 0$ is absorbing: a stopped clock has $p = 0$ and would give $r_u = 0$ forever.

Proposition 7.7. *Take a node where $\hat\eta_i = 0$, and put $f = (\bar U - k_S)/(p - k_S) < 1$, $r > r_0$. If the wheel's reference speed and acceleration have the same sign, then at rate fr the feedforward magnitude at that node is at most $\bar U$.*

Proof. With $X = k_V r|\bar v|$ and $Y = k_A r^2|a|$, the node's feedforward at rate r is $k_S + X + Y \le p$. At rate fr it is $k_S + fX + f^2 Y \le k_S + f(X + Y) \le k_S + f(p - k_S) = \bar U$. □

For a braking node, the terms of (7.3) oppose each other, and the demand is lower to begin with. The update (7.9) runs each cycle, so a single step need not be exact.

A lag limit. Let e_b be the distance by which the robot is behind the reference along the path, and $c = \max(r|v_r|, v_0)$. Then

$$r_b = 1 - \frac{[e_b - e_0]_+}{T_b c}, \tag{7.10}$$

with slack $e_0 = 2$ cm, a time $T_b = 0.15$ s, and floor $v_0 = 0.5$ m/s. The target $\min(r_u, r_b)$, clipped to $[0, 1]$, is approached at a limited rate $|\Delta r| \le a_{\max}\Delta t/c$, which bounds the extra reference acceleration $\dot r v_r$ of Proposition 7.6 by $a_{\max}|v_r|/c$.

Proposition 7.8 (Steady lag)**.** *Suppose the robot's speed is limited to $\bar v < v_r$, both constant, and r is set by* (7.10)*. Then $r \to \bar v/v_r$ and*

$$e_b \to e_0 + T_b \max(\bar v, v_0)\Big(1 - \frac{\bar v}{v_r}\Big), \tag{7.11}$$

exponentially.

Proof. We have $\dot{e}_b = r v_r - \bar{v}$. At an equilibrium $r = \bar{v}/v_r$, so $c = \max(\bar{v}, v_0)$, and (7.10) gives (7.11). With c near this value, $\dot{e}_b = v_r - \bar{v} - (v_r/(T_b c))(e_b - e_0)$ is linear with negative coefficient. □

For $\bar{v} = 1.1$ and $v_r = 1.8$ m/s this gives 8.4 cm of lag at $r = 0.61$, which a direct simulation of the rate-limited law reproduces.

If the clock lags real time by more than $T_{\max} = 2$ s, the target is set to 1, so a robot that is held still gives up rather than waiting indefinitely. Hence a motion ends within about $T_{\max}$ of its planned duration.

7.6. **The Lyapunov guard.** The solution of the quadratic program gives first-node decisions z_0, which predict speeds $y = (v_p, \omega_p)$ at the end of the first node. Let $y^\star = (v^\star, \omega^\star)$ be the RAMSETE speeds (6.5) at the current error and let $g = (g_v, g_\omega)$ be the gradient of Lemma 6.3. Set

$$f = v_{\mathrm{d}}\big(-\Phi_\varepsilon(e;\kappa)\big), \tag{7.12}$$

the rate at which $W_\varepsilon/2$ falls under RAMSETE, and fix $\nu \in (0,1)$. Let $\mathcal{T} = \{e_x^2 + e_y^2 < \tau^2,\ |e_\theta| < \tau_\theta\}$ be a tube around the path. Outside $\mathcal{T}$, the guard replaces y by

$$\Pi(y) = \begin{cases} y & \text{if } g^{\mathsf{T}}(y - y^\star) \geq -(1-\nu)[f]_+, \\ y + \dfrac{-(1-\nu)[f]_+ - g^{\mathsf{T}}(y - y^\star)}{\|g\|^2}\, g & \text{otherwise.} \end{cases} \tag{7.13}$$

A shift of $(\delta v, \delta\omega)$ in speed is a shift of $\delta v \mp \delta\omega\, \ell/2$ in wheel speeds, and a shift of γ_0 in wheel speed at the end of the first node costs one volt, so that (7.13) amounts to one correction of the two first-node voltages. This is why the rule costs almost nothing compared with an additional row in the quadratic program, and why it leaves the box structure intact.

Assumption 7.9. (i) The reference has $v_{\mathrm{d}} \geq v_{\min} > 0$ and $|\kappa| \leq \bar{\kappa}$. (ii) The errors lie in a sublevel set $\mathcal{S} = \{W_\varepsilon \leq c\}$ on which $-\Phi_\varepsilon \geq \lambda_c W_\varepsilon$ with $\lambda_c > 0$, for all $|\kappa| \leq \bar{\kappa}$ (Theorem 6.4 and Proposition 6.7). (iii) After (7.13) the voltage box is inactive. (iv) The speeds achieved are $\Pi(y) + d(t)$, with $\|d\| \leq \bar{d}$.

The term d collects wheel lag, sampling, model error, and the difference between the reference at the end of the first node, where the guard is evaluated, and the current one. The clock favors assumption (iii) by holding the feedforward near $0.95\,U$, but it remains an assumption: if the box clips the correction, the guarantee is lost for that cycle.

Theorem 7.10. (1) $\Pi(y)$ *is the Euclidean projection of* y *onto the half-plane* $\{y' : g^{\mathsf{T}}(y' - y^\star) \geq -(1-\nu)[f]_+\}$.

(2) *If* $d = 0$, *then wherever* $f \geq 0$ *and the guard is active,* $\dot{W}_\varepsilon \leq -2\nu f$. *Under Assumption 7.9, outside* $\mathcal{T}$, $\frac{\mathrm{d}}{\mathrm{d}s} W_\varepsilon \leq -2\nu\lambda_c W_\varepsilon$.

(3) *Let* $L^2 = \max(1 + 3\varepsilon^2,\, 3/b)/(1 - \varepsilon\sqrt{b})$. *Under Assumption 7.9, outside* $\mathcal{T}$,

$$\frac{\mathrm{d}}{\mathrm{d}t}\sqrt{W_\varepsilon} \leq -\nu\lambda_c v_{\mathrm{d}} \sqrt{W_\varepsilon} + L\bar{d}. \tag{7.14}$$

(4) *Consequently*

$$\limsup_{t\to\infty} \sqrt{W_\varepsilon(t)} \leq \max\Big(\sqrt{W_{\mathcal{T}}},\ \frac{L\bar{d}}{\nu\lambda_c v_{\min}}\Big), \qquad W_{\mathcal{T}} = \tau^2 + \frac{\tau_\theta^2}{b} + 2\varepsilon\tau\sin\tau_\theta, \tag{7.15}$$

provided that this bound keeps the trajectory in $\mathcal{S}$.

Proof. (1) The half-plane has outward normal $-g$. The projection of a point violating the constraint moves along g until the constraint is tight, which is (7.13).

(2) By Lemma 6.3, for applied speeds y',

$$\tfrac{1}{2}\dot{W}_\varepsilon = -f - g^{\mathsf{T}}(y' - y^\star). \tag{7.16}$$

With $y' = \Pi(y)$ the constraint gives $\frac{1}{2}\dot{W}_\varepsilon \le -f + (1-\nu)[f]_+ = -\nu f$ when $f \ge 0$. The second claim follows from $f \ge v_{\mathrm{d}}\lambda_c W_\varepsilon$ and $\mathrm{d}s = v_{\mathrm{d}}\,\mathrm{d}t$.

(3) With d present, (7.16) acquires the term $-g^{\mathsf{T}}d$, so $\frac{1}{2}\dot{W}_\varepsilon \le -\nu\lambda_c v_{\mathrm{d}} W_\varepsilon + \|g\|\bar{d}$. Now $|g_\omega| \le |e_\theta|/b + \varepsilon(|e_x| + |e_y|)$, so $\|g\|^2 \le e_x^2 + 3e_\theta^2/b^2 + 3\varepsilon^2(e_x^2 + e_y^2) \le \max(1 + 3\varepsilon^2, 3/b)N$, and $N \le W_\varepsilon/(1 - \varepsilon\sqrt{b})$ by (6.11). Divide by $\sqrt{W_\varepsilon}$.

(4) By (7.14), outside $\mathcal{T}$ the value $\sqrt{W_\varepsilon}$ falls whenever it exceeds $L\bar{d}/(\nu\lambda_c v_{\mathrm{d}})$. On $\mathcal{T}$, $W_\varepsilon \le W_{\mathcal{T}}$, directly from (6.7), whatever the plan does. □

Theorem 7.10(3) is an input-to-state stability estimate [39] for the sampled loop, in which any plan produced by the optimizer is admissible. The tube is needed because near the origin f and g are small and noise dominates. For the implemented $\tau = 1.5$ cm, $\tau_\theta = 0.05$ rad, $b = 15$, $\varepsilon = 0.1$, we have $\sqrt{W_{\mathcal{T}}} = 2.3$ cm. Proposition 6.7 gives $\lambda_c = 0.95$ on $\{W_\varepsilon \le 0.0208\}$, and $L = 1.30$. For $\nu = 0.5$, $v_{\min} = 0.5$ m/s and $\bar{d} = 0.02$ m/s the second term of (7.15) is 11 cm, which keeps the trajectory in $\mathcal{S}$, whose radius is 14 cm. The estimate is conservative: tracking errors in Section 9 are about ten times smaller.

Remark 7.11 (Scope). These guarantees are local, and we make no claim of global stability. (i) The rate in (7.14) is proportional to v_{d}. At the start and end of a path $v_{\mathrm{d}} \to 0$, and there W_ε does not increase but need not fall, in keeping with Proposition 2.3. (ii) On $\{W_\varepsilon \le 0.0873\}$, Proposition 6.7 makes W_ε fall under both the guarded plan and RAMSETE. It is therefore a common Lyapunov function for the two modes of Section 7.7, which cannot cycle inside this set, provided that the wheel-speed limit is inactive and $d = 0$. (iii) Outside that set, RAMSETE runs through a velocity loop with a speed limit. Scaling the speeds breaks (6.6), and we have no theorem there. Recovery from large errors rests on the experiments alone. (iv) All errors here are estimated errors. Section 8 treats the difference from the true error only to first order.

In the experiments of Section 9 the guard changed the plan in 1.1% of cycles. It is a safety rule that is rarely exercised, and its presence does not visibly change nominal tracking. Its effect can be seen by setting the cost to ignore errors almost entirely (q_s, q_n for 5 m, q_ψ for 50 rad, q_u for 0.5 V). The plan then does not steer, and without the guard a 10 cm cross-track error persists. With the guard, the robot converges. This experiment also shows why the strict form is required: with the function V the condition in (7.13) would be vacuous on the axis $e_x = e_\theta = 0$, and the guard would not act.

7.7. **Recovery.** The model (7.2) is a linearization, valid for small errors. If $\|(e_x, e_y)\| > \rho_f$ or $|e_\theta| > \theta_f$, with $\rho_f = 0.3$ m and $\theta_f = 0.6$ rad, the controller leaves the plan and applies RAMSETE (6.5) through a velocity loop with feedforward, and keeps $r \ge 0.5$, since (6.5) needs $v_{\mathrm{d}} \ne 0$ to correct cross-track error. It returns to the plan once both errors are below 0.7 of these limits. This hysteresis excludes chattering: since the error rates are bounded by the speed limits, a return and a re-entry are separated in time by at least $\min(0.3\rho_f/(v_{\max} + |v_{\mathrm{d}}|),\ 0.3\theta_f/(\omega_{\max} + |\omega_{\mathrm{d}}|))$, so there are finitely many switches on any bounded interval. A pose that is not a number, or a state outside any meaningful range, resets the solver and outputs zero volts for that cycle.

7.8. **Learning a steady loss.** Let a wheel lose an unknown voltage η, as from a rubbing gear, so that the plant is $\dot{v} = (u - \eta - k_S \operatorname{sgn} v - k_V v)/k_A$. At each cycle of length h the speed is

compared with a one-step prediction made with the estimate $\hat{\eta}$ and the command u_- to which the delayed speed measurement responds:

$$\nu = v^+ - \Big(v + \frac{h}{k_A}\big(u_- - \hat{\eta} - k_S \operatorname{sgn} v - k_V v\big)\Big), \qquad \hat{\eta}^+ = \hat{\eta} - \frac{k_A}{T_\ell}\nu, \tag{7.17}$$

with $\hat{\eta}$ clipped to ± 3 V. The update is applied only when the wheel's speed exceeds 0.15 m/s, so that it is out of stiction, and when both wheels are cruising, with acceleration below 0.75 m/s^2.

Proposition 7.12. *Let the plant be the Euler model with period h.*

(1) *The error $\tilde{\eta} = \hat{\eta} - \eta$ obeys $\tilde{\eta}^+ = (1 - h/T_\ell)\tilde{\eta}$.*
(2) *For the exact plant of Lemma 3.3, the update has a fixed point with $\tilde{\eta} \approx \frac{h}{2T_w}(u_- - \eta - k_S \operatorname{sgn} v - k_V v)$. This is the voltage that produces the acceleration, multiplied by $h/(2T_w)$, which is why updates are limited to cruising.*
(3) *If the measured speed carries white noise of variance σ^2, the stationary standard deviation of $\hat{\eta}$ is*

$$\frac{k_A}{T_\ell}\,\sigma\sqrt{\frac{1 + c^2 - 2ac}{1 - a^2}}, \qquad a = 1 - \frac{h}{T_\ell}, \quad c = 1 - \frac{h}{T_w}, \tag{7.18}$$

which is approximately $(k_A/T_\ell)\sigma\sqrt{1 + hT_\ell/(2T_w^2)}$.

Proof. (1) Substituting the plant in (7.17) gives $\nu = (h/k_A)\tilde{\eta}$.

(2) For the exact plant the true increment is $\frac{1-\alpha}{k_V}\Theta$ with $\Theta = u_- - \eta - k_S \operatorname{sgn} v - k_V v$, and $(1 - \alpha)/k_V = (h/k_A)(1 - h/(2T_w) + O(h^2))$. Thus $\nu = (h/k_A)(\tilde{\eta} - \frac{h}{2T_w}\Theta) + O(h^3)$, which vanishes at the stated value.

(3) With noise n_k on the measurements, ν_k contains $n_{k+1} - (1 - h/T_w)n_k$, up to terms already counted, so $\tilde{\eta}$ is an ARMA$(1,1)$ process with autoregressive coefficient a and moving-average coefficient c, driven by noise of standard deviation $(k_A/T_\ell)\sigma$. Its variance is the standard formula, and the approximation follows from $1 + c^2 - 2ac = (1 - c)^2 + 2c(1 - a)$. □

With $T_\ell = 0.5$ s, $T_w = 0.095$ s and $\sigma = 0.01$ m/s the noise floor is about 13 mV. The term $\hat{\eta}$ is faded out below $v_\ell = 0.5$ m/s of reference speed in (7.3), because the model of a constant loss is no longer meaningful near stiction.

7.9. **One control cycle.** Algorithm 2 assembles the pieces. Before the error is computed, the pose and wheel speeds are carried forward by the latency $D = 15$ ms from a pose reading to the action of the motors, using the measured speeds and the previous command.

Table 1 counts floating-point operations. The largest cost is forming H, and it grows with the square of the number of variables in each of N stages. This is why the number of nodes matters more than the solver. The experiments show that five nodes track as well as eight on all the benchmark conditions, because the terminal cost carries most of the information about later time. Eight nodes held a path more closely at severe voltage limits (0.5 cm against 1.5 cm at 7 V). All storage is fixed, with an object of 4.8 kB.

8. The closed loop with the estimator

8.1. **Combined architecture.** Figure 6 shows the system. The controller acts on the estimated error, and the estimator does not know what the controller intends. To first order the two errors add.

Algorithm 2 One cycle of the predictive tracker

$\tau \leftarrow \tau + r\Delta t$; update $\hat{\eta}$ by (7.17)
sample the reference at $\tau + r(D + \sum_{j<k} h_j)$, $k = 0, \dots, N$; form (7.3), the box (7.5), and the peak p
carry pose and wheel speeds forward by D; form x_0 and the body-frame error e
$r \leftarrow$ rate-limited step toward $\mathrm{clip}(\min(r_u, r_b), 0, 1)$; set the target to 1 if $t - \tau > T_{\max}$
if the error is not a number **then** reset the solver; **return** 0
end if
if the error is out of range (with hysteresis) **then return** RAMSETE through a velocity loop
end if
build H, g by (7.6), with P from the table
$z \leftarrow$ Algorithm 1, warm-started, at most 6 iterations
apply the guard (7.13) to the first node, and clip to the box
return $u^{\mathrm{ff}}_{i,0} + \delta_{i,0}$

TABLE 1. Floating-point operations per cycle, when one solver iteration suffices.

Operation	$N = 5$	$N = 8$	Scaling
Rollout of state and responses	575	1472	$23N^2$
Input responses	120	192	$24N$
Stage cost in H and g	1700	5520	$O(N^3)$
Terminal cost in H and g	1150	2320	$O(N^2)$
Reference and feedforward	380	584	$O(N)$
Cholesky factorization	333	1365	$(2N)^3/3$
Triangular solves	200	512	$2(2N)^2$
Interpolation of P	75	75	constant
Total	4533	12040	

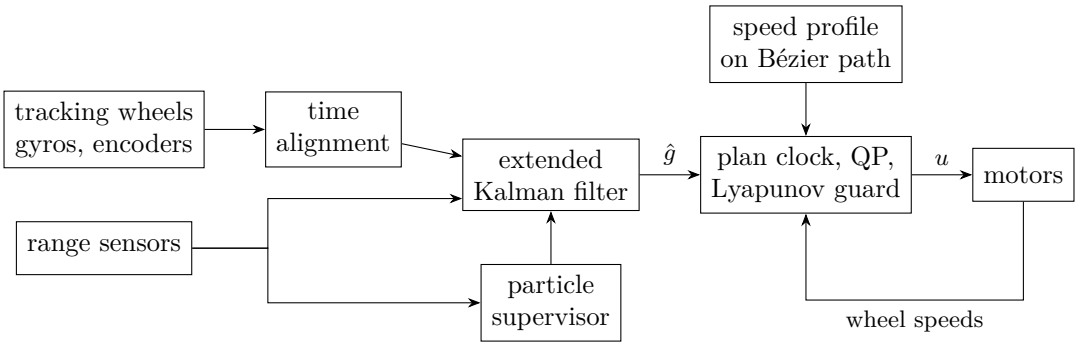


FIGURE 6. The estimator and the controller. The estimator runs every 5 ms and the controller every 10 ms.

Proposition 8.1. *Let the true error at the end of a motion be $\varepsilon_c + \varepsilon_l$, where ε_c is the error of the controller relative to the estimate and ε_l the error of the estimate, uncorrelated and of zero mean, with position covariances Σ_c, Σ_l. Then $\lambda_{\max}(\Sigma_c + \Sigma_l) \le \lambda_{\max}(\Sigma_c) + \lambda_{\max}(\Sigma_l)$. So if the*

two parts have 3σ radii B_c and B_l in the sense of Proposition 4.10, the total has radius at most $(B_c^2 + B_l^2)^{1/2}$.

Proof. The largest eigenvalue is subadditive on symmetric matrices, and the radius is proportional to the square root of the largest eigenvalue. □

The implementation reports this quantity with each motion, with the settling tolerance standing in for B_c, as a prediction of final error made before the motion ends. If $\hat{e}$ is the error seen by the controller, the ultimate bound (7.15) holds for $\hat{e}$. Since $e = \hat{e} + O(\|\varepsilon_l\|)$ the true error is bounded by the sum, so better control cannot pay for bad localization. Table 4 shows the effect as the calibration worsens.

8.2. **Ending a path.** After the clock reaches the end of the reference the implementation switches to a law that closes the distance along the final heading and turns onto that heading, with $e_\parallel$ and e_ψ the errors along and about that heading:

$$v = \mathrm{sat}_{0.5}(3e_\parallel), \qquad \omega = \mathrm{sat}_3(6e_\psi). \tag{8.1}$$

Sideways error is left alone. By Proposition 2.3, to remove a sideways error d while staying within ψ of the final heading requires a path of length at least $d/\sin\psi$, which is 5.8 cm of travel per centimeter of correction at $\psi = 10^\circ$. A differential drive at rest can do this only by a maneuver, and by Brockett's theorem no continuous time-invariant feedback stabilizes the pose.

The motion is declared finished when the distance to the end is below 1.9 cm, or when $|e_\parallel|$ is below 1.9 cm and $|e_\psi|$ below 2°. Since the first test does not examine the heading, and since the plan usually arrives within that distance, the law (8.1) typically runs for about 8 ms. With the predictive tracker this does little harm, because the heading is already right on arrival. It does mean that the exit does not itself correct anything, and Section 9.8 quantifies what follows.

9. Numerical experiments

9.1. **Setup.** All results are from simulation. The simulator integrates a rigid body at 5 kHz with six tire contacts, friction circles and load transfer, wall contact by penalty forces, and a motor model with back-EMF, a 2.5 A current limit, friction and battery sag. Motor commands are delayed by 5 ms. Tracking wheels, gyros, range sensors and drive encoders run on their own clocks with quantization, noise, latency and per-unit calibration error. The range sensors, of which there are five, have a beam cone, noise that grows with range and incidence, spurious returns and dropouts. Unmapped objects block beams. The estimators and controllers are compiled from the same source that runs on the robot. Four calibration grades, called ideal, careful, typical and rushed, set the residual calibration errors and feedforward errors, the last with a standard deviation of 0, 2, 6 and 12%. Most parameters are informed estimates and have not been fitted to data from a robot. The manufacturer quotes the accuracy of the distance sensor as about ±15 mm below 200 mm and about 5% above [43], while the simulator's standard deviation at normal incidence is 0.4 cm plus 0.8% of the range, or 1.2 cm at 1 m. Whether the quoted figure means one or two standard deviations is not stated, so the model may be optimistic by a factor of two to four. The simulator also shares structure with what it tests: its motors follow (3.2) closely, which suits the predictive model, and its sensors use the same beam cone as the estimator. Gear backlash and compliance are not modeled. For these reasons the results below may favor the methods studied here, and Section 9.4 measures part of that dependence.

TABLE 2. Localization error over 40 runs per scenario. The push scenario shoves the robot 0.5 m with its tracking wheels off the ground, so its RMS and consistency are dominated by the second or so before the sensors relocalize; its final error is the figure of interest.

	RMS error (cm), median			Filter final	Filter	Inside
Scenario	typical	exact	filter	95th pct. (cm)	NEES	3σ
Skills, 50 s	14.35	4.63	0.47	0.58	2.90	99.6%
Autonomous, 15 s	1.60	1.39	0.36	0.47	2.45	99.1%
Aggressive driving	12.79	2.73	0.58	0.46	4.97	98.9%
Push	75.65	75.72	24.44	0.41	121	75.9%

TABLE 3. Ablation of the filter on the skills scenario.

Filter variant	RMS (cm)	RMS 95th pct. (cm)	NEES	Inside 3σ
All features	0.47	0.58	2.90	99.6%
No time or latency alignment	2.55	2.78	89.2	50.9%
No self-calibration	0.56	0.97	24.4	94.0%
No redundant wheels	0.53	1.90	9.14	97.6%
No zero-velocity updates	0.47	0.61	14.0	99.6%

9.2. **Localization.** Table 2 reports 40 Monte Carlo runs for each of four scenarios, in which a scripted driver steers on the true pose so that all estimators see identical motion. "Typical odometry" is the common practice of integrating the newest value of each sensor with one gyro and nominal calibration. "Exact odometry" uses Lemma 2.1 and time alignment with three gyros. Figure 7 shows one skills run, with a filter RMS of 0.43 cm and 99.8% of samples inside the filter's own 3σ radius. Table 3 removes one feature at a time. Time alignment is the largest single factor, as Proposition 4.1 predicts, and the increase of NEES without self-calibration is the effect of Proposition 4.8.

9.3. **Tracking.** Here the estimator is in the loop on the same simulator. Four paths (an S-curve, a hairpin, a long sweep and a backward S-curve) are each run with ten seeds in each of eight conditions. The first four are the calibration grades. In the last four the careful robot is 25% heavier than its characterization, has an 11.4 V battery, loses 1.5 V on one side, or is pushed sideways with 30 N for 0.25 s at the middle of the path. Table 4 gives means over the 40 runs, and Figure 8 the medians and spreads. The score is the true pose. Distance from the end is measured 0.3 s after the motion finishes.

With ideal calibration the predictive tracker reduces RMS tracking error by a factor of 2.5 against RAMSETE and of 3.4 against LTV. As calibration worsens the gap narrows, since more of the error belongs to the estimator (Proposition 8.1). The largest gains come where the model is wrong in ways the tracker can see: a dragging side, where it is three times better on the path and nearly four times better at the end, and extra mass. On the two whole-routine scores in the typical condition, the median and worst miss over all motions went from 1.58 cm and 4.43 cm to 1.13 cm and 3.50 cm for an 8-second routine, and from 0.83 cm and 2.76 cm to 0.71 cm and 2.30 cm for a 49-second routine. Durations match the plan within 10 ms on average.

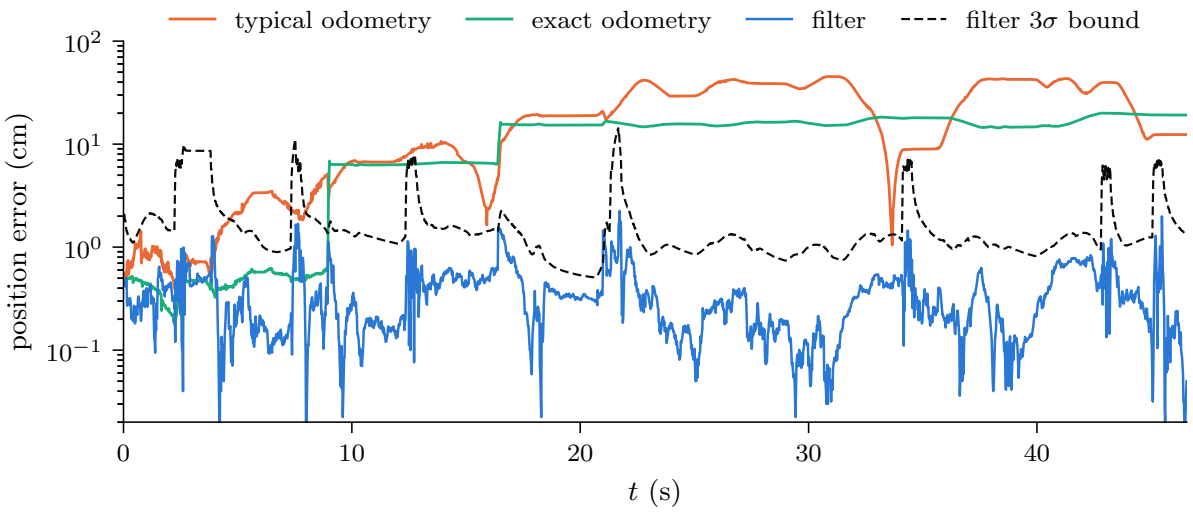


FIGURE 7. Position error on one skills run (seed 1), on a logarithmic scale. The bound rises where few beams are usable.

TABLE 4. RMS distance from the path / distance from the end point, in cm, over 40 runs per row. The smallest in each column pair is in bold.

	RAMSETE		LTV		Predictive	
Condition	path	end	path	end	path	end
Ideal calibration	1.30	0.97	1.74	0.87	**0.51**	**0.50**
Careful team	1.33	0.91	1.77	0.99	**0.65**	**0.52**
Typical team	1.64	1.29	2.07	1.24	**1.12**	**0.97**
Rushed team	2.51	1.78	3.08	2.23	**2.11**	**1.30**
25% heavier than characterized	1.50	1.75	1.92	1.11	**0.67**	**0.56**
Tired battery (11.4 V)	1.32	0.92	1.87	1.13	**0.64**	**0.55**
Left side loses 1.5 V	2.34	2.64	3.29	3.06	**0.73**	**0.70**
Shoved sideways mid-path	1.60	0.96	1.77	1.05	**1.00**	**0.63**

9.4. **Sensitivity to the simulator.** A comparison run on one's own simulator can favor one's own model. To see how far the ranking depends on assumed parameters, we repeat the benchmark for the careful team with one assumption changed at a time, well past its default. The last row changes several at once: range noise ×2, gyro noise ×3, grip ×0.7, omni-wheel scrub ×2, resistance ×1.3, a command delay of 15 ms and a speed filter of eight samples. Each change is made to the plant alone. The feedforward, the estimator's noise model and the controller's latency keep their defaults, so each change is an unmodeled mismatch. Table 5 gives the results. A range noise of ×2 corresponds to reading the manufacturer's 5% as two standard deviations, and ×4 to reading it as one.

The predictive tracker keeps the lowest mean path error in every row. Its end-point advantage is less secure. With range noise ×4 it wins 29 of 40 runs on the path, and the end errors are nearly level, at 1.45 cm against 1.50 cm, with the estimate itself degraded to 1.6 cm. With a speed filter of twice the length, RAMSETE ends closer, at 0.83 cm against 1.10 cm, and the

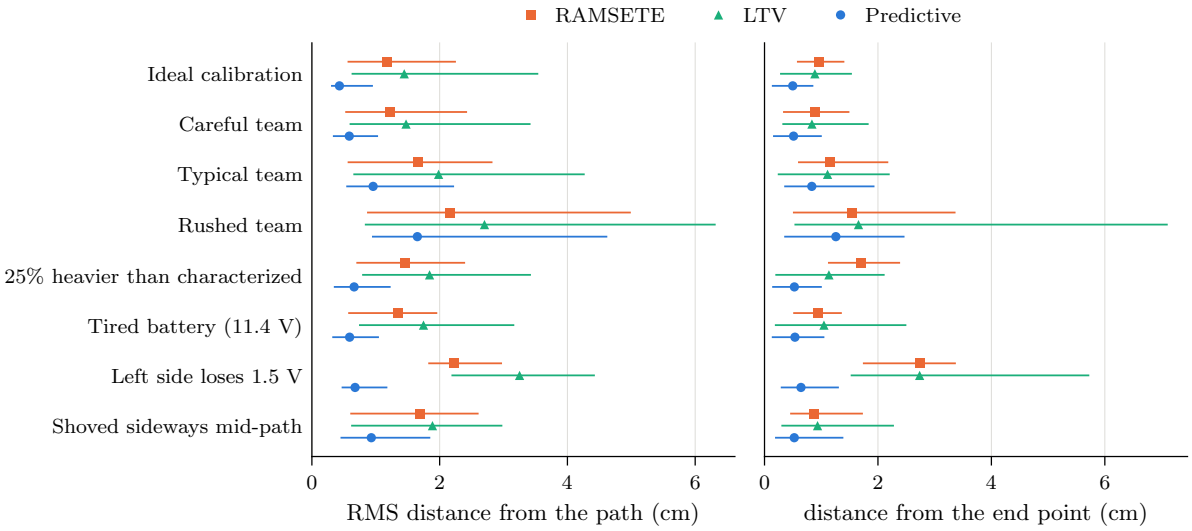


FIGURE 8. Median and 5th to 95th percentile over the 40 runs of each condition.

TABLE 5. Sensitivity of the comparison, 40 runs per row. Columns give mean RMS distance from the path / mean distance from the end, in cm. "Wins" counts the paired runs of 40 in which the predictive tracker beat RAMSETE on each measure. "Est." is the RMS error of the estimate.

	RAMSETE		LTV		Predictive		Wins		Est.
Change	path	end	path	end	path	end	path	end	(cm)
Baseline (careful team)	1.33	0.91	1.77	0.99	0.65	0.52	39	33	0.85
Range noise ×2	1.35	0.95	1.80	1.00	0.71	0.71	39	28	1.03
Range noise ×4	1.63	1.50	2.05	1.84	1.03	1.45	29	22	1.57
Gyro noise ×3	1.32	0.93	1.79	0.97	0.65	0.54	40	33	0.87
Tire grip ×0.7	2.79	2.18	3.60	3.01	1.88	0.87	40	38	0.93
Tire grip ×1.3	1.24	0.89	1.45	0.90	0.55	0.57	40	32	0.88
Side scrub ×4	2.16	2.04	2.95	2.44	0.84	0.70	40	40	0.87
Motor resistance ×1.3	1.55	1.94	1.98	1.20	0.68	0.62	40	39	0.89
Command delay 20 ms	1.41	1.36	1.85	1.06	0.66	0.56	40	37	0.88
Speed filter twice as long	1.27	0.83	1.72	0.94	0.59	1.10	40	14	0.85
All together	3.16	3.27	4.12	3.90	1.73	1.14	40	38	1.10

predictive tracker wins only 14 of 40 runs on that measure. The plan starts from measured speed and its latency correction does not know about the added lag, which is the likely cause, though we have not tested that explanation. Lower grip raises the errors of every tracker, and the predictive tracker keeps its lead at the end point there. These tests cover the listed parameters only, and they do not replace identification on hardware.

9.5. **Saturation.** To isolate the effect of the voltage limit, we use the exact first-order plant of Section 3 with $k_S = 0.7$, $k_V = 5.5$, $k_A = 0.7$, a path planned at 1.8 m/s, and a limit of 7 V, which is 58% of what the plan assumes. Figure 9 shows the result. RAMSETE with a per-wheel clamp leaves the path by 10.9 cm. When its time-indexed reference ends, it stops 91 cm short,

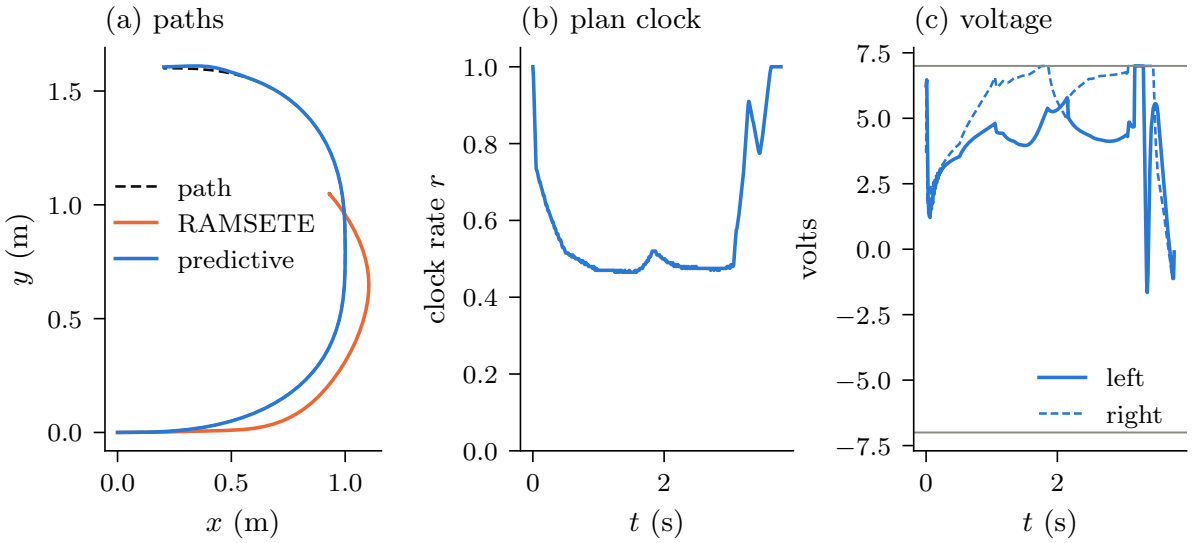


FIGURE 9. A path planned at 1.8 m/s, driven at a 7 V limit. (a) Paths. The predictive trace covers the drawn path. (b) The clock rate. (c) Its wheel voltages, with the limit shown. The clock keeps the feedforward near 0.95 of the limit, which leaves room for feedback.

since (6.5) has $k = 0$ for a stopped reference. The predictive tracker stays within 1.5 cm of the path and reaches its end 0.65 cm off, by running the clock at $r \approx 0.47$ and arriving 1.6 s late.

9.6. **Solver and guard.** Over all 119,234 control cycles of the predictive runs, the solver averaged 1.01 iterations per cycle and at most 3, and it never reached its limit of 6. The guard acted in 1.1% of cycles. The recovery law never engaged in these conditions; separate tests start the robot 45 cm off the path with a heading error of 1.2 rad, and it returns to the path.

9.7. **Cost on the target processor.** The V5 brain has a Cortex-A9. The manufacturer lists the system processor at 667 MHz and the user processor at 1333 MIPS [44]. We have not run this code on a brain. Instead we compiled the controller and estimator with the firmware's compiler and flags and ran them on an emulated Cortex-A9 with instruction counting, which gives exact dynamic instruction counts for the compiled code but no timing. To convert counts to time we measured the instruction mix and applied the double-precision throughput and latency of the processor's floating-point unit [3]. A lower estimate takes floating-point operations at their throughput and all other instructions at one cycle; an upper estimate takes them at their full latency and others at 1.5 cycles. This gives 1.18 to 2.53 cycles per instruction for the controller and 1.28 to 2.53 for the estimator, at an assumed 667 MHz. These figures omit cache misses, operating-system overhead and device reads, so they are estimates, not bounds. The estimator workload has five range sensors, each read every seventh tick, or 35 ms, and staggered so that five of every seven ticks carry one reading.

Table 6 gives the results. The controller is inexpensive, at under 1% of its period by this estimate, and about 1.4 million instructions at startup to build its tables. The estimator is not. With 300 particles, averaged over ticks, it may take 48 to 95% of the processor, and a tick with a range reading may overrun its 5 ms period. With no particles the same average is 11 to 21%. The particle filter accounts for about 90% of a tick without a range reading and 76% of a tick with one. It is needed only for recovery after a collision or a loss of position, so a lower count

TABLE 6. Instructions per call on an emulated Cortex-A9, and the estimated time at 667 MHz, with its share of the period.

Workload	Mean	Worst	Time (ms)	Share
Controller, 10 ms period	14,800	18,100	0.026–0.056	0.3–0.6%
Estimator, 5 ms period, 300 particles:				
tick without a range reading	494,900	500,200	0.95–1.88	19–38%
tick with a range reading	1,565,500	1,710,100	3.00–5.94	60–119%
100 particles:				
tick without a range reading	198,400	201,700	0.38–0.75	8–15%
tick with a range reading	773,500	841,000	1.48–2.93	30–59%
Filter only, no particles:				
tick without a range reading	50,200	53,100	0.10–0.19	2–4%
tick with a range reading	376,600	420,600	0.72–1.43	14–29%

TABLE 7. Errors at the end of a path, median / 95th percentile, over 40 runs each. The last column is the 95th percentile of the sideways miss after a further 61 cm straight ahead. "Shoved at the end" applies 30 N sideways for 0.25 s, starting 0.45 s before the end.

		Along (cm)		Sideways (cm)		Heading (°)		Miss at
Condition	Tracker	med	95%	med	95%	med	95%	61 cm
Careful team	RAMSETE	0.47	1.07	0.55	1.34	2.09	4.12	4.97
Careful team	Predictive	0.31	0.75	0.25	0.71	0.16	0.34	0.76
Typical team	RAMSETE	0.69	1.71	0.82	1.61	1.29	4.57	5.22
Typical team	Predictive	0.50	1.11	0.55	1.70	0.18	0.81	1.91
Rushed team	RAMSETE	0.90	2.39	0.96	2.90	0.73	4.53	5.30
Rushed team	Predictive	0.70	2.02	0.90	2.09	0.37	0.91	2.47
Shoved at the end	RAMSETE	1.09	2.11	2.66	20.20	0.88	2.44	19.51
Shoved at the end	Predictive	0.32	0.79	3.19	19.14	0.64	0.96	19.61

or a lower rate may be enough, but we have not measured what that costs in recovery. Until these numbers are confirmed on a brain, the estimator is the main real-time risk in this design.

9.8. **How paths end.** Table 7 gives the error at the end of a path resolved into its three parts, and the sideways miss that would result from driving straight on for 61 cm (24 in), as one does when lining up on a goal. On clean runs the predictive tracker arrives with almost no heading error, which matters more than position for the last row: its miss at 61 cm is less than half that of RAMSETE.

The last two rows show the limitation of Section 8.2. After a late shove, heading and along-track error remain small but the robot ends up to 19 cm to the side. Its own estimate was within 0.74 cm of the truth at the 95th percentile, so the robot knows. Neither tracker can act on this in the distance remaining, by Proposition 2.3. A planned re-approach, which backs up along the final heading and drives in again, would be needed.

10. Conclusions and open problems

We have developed the mathematics of a complete localization and tracking system for a voltage-limited differential drive. The estimation results explain, in closed form, why clock skew matters and how to read observability from the geometry of a sensor layout. The control results give a strict certificate for a widely used tracking law, an exact and finite method for its predictive extension, and, under the stated assumptions, a local guarantee that holds for any plan the optimizer returns. The experiments favor the design on path tracking across the conditions tested, with the qualifications of Sections 9.4 and 9.7.

Several questions remain open.

(1) *Hardware.* No result here has been tested on a robot. The first step is identification: stationary logs for sensor noise, ramp and step tests for the feedforward constants and delays, and a measured speed-filter lag, followed by repeating Tables 4 and 5 with fitted values.
(2) *Estimator cost.* Table 6 is an estimate from instruction counts. The time on a brain must be measured, and the particle supervisor must be made cheaper without losing recovery.
(3) *Sensitivity of the plan to speed lag.* Table 5 shows that a slow speed estimate hurts the end error of the predictive tracker. An observer for wheel speed, or a latency matched to the measured lag, should be tried.
(4) *The optimal closed loop.* Theorem 7.10 treats the plan as an arbitrary input and says nothing about the improvement the optimizer provides. A proof that the non-uniform grid with a terminal cost computed at a different step retains the usual stability properties of receding-horizon control would be of interest.
(5) *Stability.* Remark 7.11 lists what is not covered. A result that spans the speed-limited recovery mode, and a sharper ultimate bound, are open.
(6) *Coupling with the estimator.* Proposition 8.1 treats the errors as uncorrelated. The controller's commands excite the estimator's calibration states, and the true coupling has not been analyzed.
(7) *Joint optimization of timing.* The clock of Section 7.5 is a feedback rule. Treating the clock rate as a decision variable would make the problem nonconvex in the present form, but the change of variables of [42] may give a tractable version.
(8) *Ending a path.* Proposition 2.3 bounds what any maneuver must do, and the re-approach suggested in Section 9.8 should be designed and analyzed, along with a stricter exit test.

Appendix A. Jacobians of the motion model

Let $\Delta = (\Delta x, \Delta y)^{\mathsf{T}}$, $\phi = \Delta\theta$, $\mathrm{V} = \mathrm{V}(\phi)$ and $\mathrm{V}' = \partial_\phi \mathrm{V}(\phi)$. From (4.2),

$$\frac{\partial p^+}{\partial \theta} = \mathrm{J}\mathrm{R}(\theta)\mathrm{V}\Delta, \qquad \frac{\partial p^+}{\partial \Delta} = \mathrm{R}(\theta)\mathrm{V}, \qquad \frac{\partial p^+}{\partial \phi} = \mathrm{R}(\theta)\mathrm{V}'\Delta, \qquad \frac{\partial \theta^+}{\partial \phi} = 1.$$

By (4.1), with $S = 1 + s_g$,

$$\frac{\partial \phi}{\partial b_g} = -\frac{h}{S}, \quad \frac{\partial \phi}{\partial s_g} = -\frac{\phi}{S}, \quad \frac{\partial \Delta x}{\partial \mu_\parallel} = r_\parallel \Delta\phi_\parallel, \quad \frac{\partial \Delta y}{\partial \mu_\perp} = r_\perp \Delta\phi_\perp, \quad \frac{\partial \Delta x}{\partial o_\parallel} = \phi, \quad \frac{\partial \Delta y}{\partial o_\perp} = -\phi,$$

and $\partial\Delta x/\partial\phi = o_\parallel$, $\partial\Delta y/\partial\phi = -o_\perp$. The chain rule assembles F. The implementation computes these by central differences, and the two agree to 10^{-9}.

Appendix B. Parameters

Table 8. Values used in the implementation and the experiments.

Symbol	Value	Symbol	Value
m, I	6.8 kg, 0.20 kg m^2	ℓ	0.33 m
G, r_w	4/3, 41.3 mm	Motors per side	3
K_e, K_t, R	0.191, 0.14, 2.0	$i_{\max}$	2.5 A
k_S, k_V, k_A	0.27, 6.31, 0.60	T_w	0.095 s
$v_{\max}$, $a_{\max}$	1.8 m/s, 3 m/s^2	a_c	6 m/s^2
Ω, Λ	9 rad/s, 30 rad/s^2	U	12 V
b, ζ	15, 0.7	ε, ν	0.1, 0.5
τ, τ_θ	1.5 cm, 0.05 rad	ρ_f, θ_f	0.3 m, 0.6 rad
q_s, q_n, q_ψ from	3 cm, 2 cm, 0.1 rad	q_w, q_u from	0.5 m/s, 6 V
N, $k_{\max}$, n_p	5, 6, 3	D, h_T	15 ms, 0.05 s
e_0, T_b, v_0	2 cm, 0.15 s, 0.5 m/s	r_0, $T_{\max}$	0.05, 2 s
T_ℓ, v_ℓ	0.5 s, 0.5 m/s	Limit on $\hat{\eta}$	3 V
N_g	2.6×10^{-4} rad/s/$\sqrt{\text{Hz}}$	q_g	10^{-5} rad
Slip rate	9×10^{-6} m^2/m	Particles	300
Filter period	5 ms	Control period	10 ms

Acknowledgment

This paper is the culmination of the author's work in competition robotics. Its problems and requirements came from the author's experience programming VEX robots and from the author's earlier robot code. The author decided to combine predictive control and RAMSETE tracking into one controller, required that the result be cheap enough to run on the V5 brain, directed the project at each step, and tested and challenged the results throughout. The derivations, software and text were developed with Claude, an AI system by Anthropic, under the author's direction. The proofs have not been independently reviewed.

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